

PRIMARY 1 BASIC SCIENCE SCHEME OF WORK

FIRST TERM

Week	Topic	Breakdown (Subtopics)
1	Exploring Our Environment (Part 1)	• Understanding what an environment is • Identifying our immediate surroundings (home, school, classroom) • Naming things we see in our environment • Describing our environment using our senses (what we see, hear, smell, touch, taste) • Taking a nature walk around the school compound
2	Exploring Our Environment (Part 2)	• Exploring the school environment: classrooms, playground, toilets, staff room • Exploring the home environment: bedroom, kitchen, living room, bathroom, compound • Identifying people in our environment: family, teachers, friends, neighbours • Understanding safety in our environment • Taking care of our environment (keeping it clean)
3	Exploring Other Parts of the Surrounding: Roads	• Identifying roads in our community • Types of roads: tarred roads, untarred roads, pathways • Uses of roads: for vehicles, walking, bicycles • Road safety: crossing the road safely, traffic lights (red, yellow, green) • People who work on roads: drivers, road

		safety officials, road construction workers
4	Colour Identification	- Identifying primary colours: red, blue, yellow - Identifying secondary colours: green, orange, purple - Identifying other common colours: black, white, brown, pink, grey - Colours in our environment: colour of sky, grass, sun, etc. - Sorting and grouping objects by colour - Mixing colours (practical activity with paints/crayons)
5	MID-TERM EXAMINATION	Assessment of Weeks 1–4
6	MID-TERM BREAK	Holiday/Rest
7	Shapes (Part 1)	- Identifying basic shapes: circle, square, triangle, rectangle - Properties of shapes: sides and corners - Finding shapes in our environment (windows, doors, wheels, balls) - Drawing and tracing basic shapes - Sorting objects by their shapes
8	Shapes (Part 2)	- Identifying more shapes: oval, diamond, star, heart - Comparing shapes: big and small, same and different - Creating patterns with shapes - Using shapes to make pictures and designs - Three-dimensional shapes: ball (sphere), box (cube), tin (cylinder)
9	Living Things (Part 1)	Understanding what living things are

		• Characteristics of living things: they move, eat, grow, breathe, reproduce • Examples of living things: humans, animals, plants • Identifying living things in our environment • Differences between different living things
10	Living Things (Part 2)	• Animals as living things: pets (dog, cat, rabbit), farm animals (goat, cow, chicken), wild animals (lion, elephant, monkey) • Plants as living things: trees, flowers, grasses, vegetables • Parts of plants: roots, stem, leaves, flowers, fruits • What living things need to survive: food, water, air, shelter • Taking care of living things (plants and animals)
11	REVISION	• Review of all topics: Environment, roads, colours, shapes, living things
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends

PRIMARY 1 BASIC SCIENCE LESSON NOTES

FIRST TERM

WEEK 1: EXPLORING OUR ENVIRONMENT (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define environment simply as everything around us.
2. List three things found in the classroom.
3. Mention the five senses used to explore the environment.
4. Identify objects based on what they see, hear, or touch.

CONTENT



1. Understanding Environment

- The environment is everything we see, hear, feel, smell, and taste around us. It is our surroundings.
- It includes the place where we live, the school where we learn, and the playground where we play. It is our home, our street, and everywhere we go.
- Everything around us—people, trees, buildings, animals, air, and water—is part of our environment. Even things we cannot see, like air, are part of it.

2. Identifying Surroundings

- **In the Classroom:** We see tables, chairs, the chalkboard, books, pencils, our teacher, and our friends. We also feel the smooth desk and hear the teacher's voice.
- **In the Home:** We see beds, televisions, pots, plates, clothes, and our family members. We smell food from the kitchen.
- **Outside:** We see trees, sand, grass, cars, birds, the sky, clouds, and the sun. We feel the wind on our skin.

3. Describing Environment Using Senses

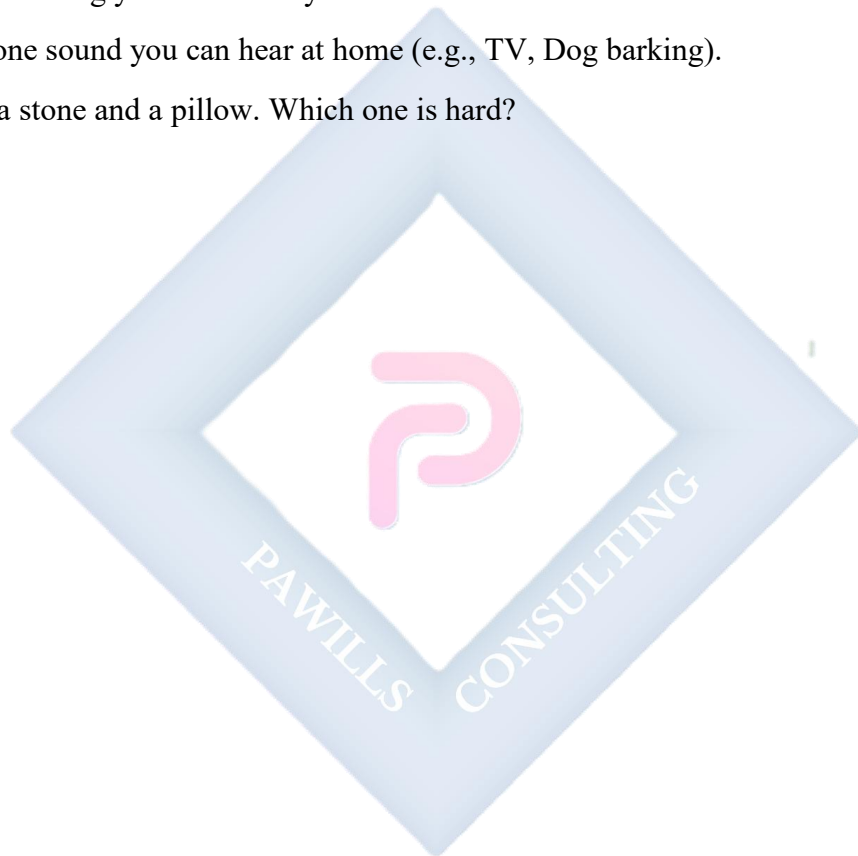
- We have five special ways to know our environment. These are our senses. They are like tools our body uses to learn about the world.
 - **Eyes (Sight):** We use our eyes to see colors, shapes, sizes, and movement. Example: We see a red ball rolling.
 - **Ears (Hearing):** We use our ears to hear sounds like a bell ringing, a bird singing, our name being called, or music playing.
 - **Nose (Smell):** We use our nose to smell good food, flowers, soap, or sometimes bad smells like rubbish.
 - **Tongue (Taste):** We use our tongue to taste if something is sweet (like sugar), sour (like lemon), salty (like salt), or bitter (like bitter leaf).
 - **Skin (Touch):** We use our hands and skin to feel if something is hot, cold, soft, hard, rough, or smooth. Example: A pillow is soft, a stone is hard.

EVALUATION

1. What is the environment?
2. Name two things you can see in your classroom.
3. Which part of the body do we use to see?
4. Which part of the body do we use to smell?
5. Mention one thing you can touch in the school compound.

ASSIGNMENT

1. Draw one thing you can see in your classroom.
2. Name one sound you can hear at home (e.g., TV, Dog barking).
3. Touch a stone and a pillow. Which one is hard?



WEEK 2: EXPLORING OUR ENVIRONMENT (PART 2)

OBJECTIVES

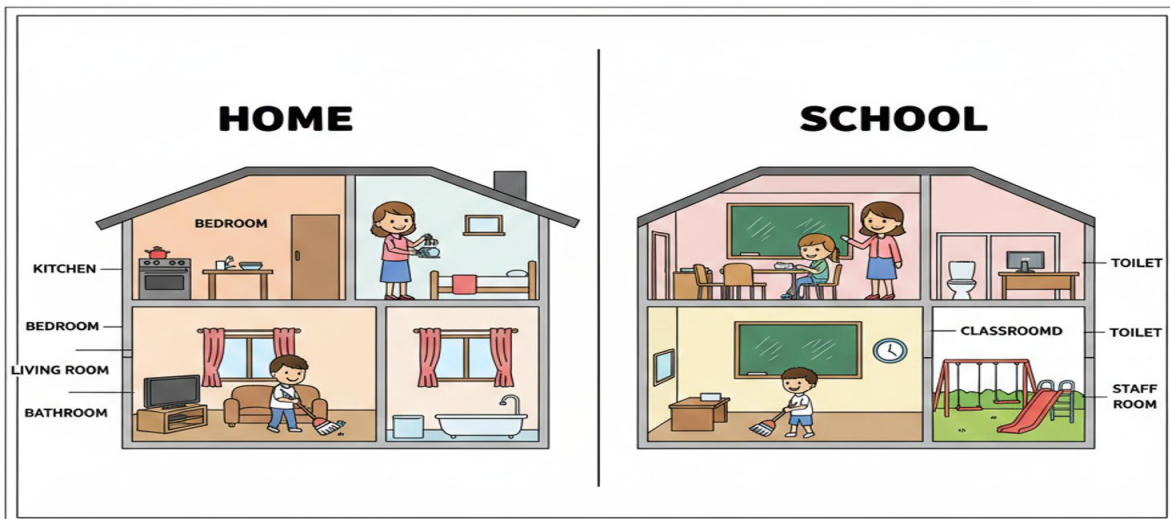
By the end of the lesson, pupils should be able to:

1. Identify different rooms in the home and their uses.
2. Identify key areas in the school environment.
3. Mention three groups of people found in our environment.
4. State one way to keep the environment clean.

CONTENT

1. Exploring the School Environment

- **Classroom:** This is where we sit to learn, write, read, and listen to the teacher. It has desks, a board, and learning charts.
- **Playground:** This is an open space where we run, jump, play football, and swing during break time. It helps us exercise and have fun.
- **Toilets:** This is where we go to ease ourselves (urinate and defecate). We must keep it clean by flushing and washing our hands.
- **Staff Room:** This is where the teachers sit, plan lessons, mark books, and rest. It is their office.



2. Exploring the Home Environment

- **Living Room (Parlour):** We sit here to relax, watch television, welcome visitors, and spend time with family.
- **Kitchen:** This is where Mummy or Daddy cooks food. It has a stove, pots, plates, and food items. We must be careful here.
- **Bedroom:** This is where we sleep at night on a bed. We keep our clothes here and it is a quiet place for rest.
- **Bathroom:** This is where we take a bath to clean our bodies. It has water, soap, and a towel.

3. People in Our Environment

- **Family:** Father, Mother, Brother, Sister, Grandparents. They love and care for us at home.
- **School:** Teachers (teach us), Headmistress/Headmaster (leads the school), Classmates (learn with us), Cleaners (keep the school clean).
- **Community:** Neighbours (live near us), Doctors (make us well when sick), Police officers (keep us safe), Traders (sell things).

4. Safety and Care

- We must not play with sharp objects like knives, blades, or broken glass because they can cut us.
- We must not play near fire in the kitchen or with matches. Fire can burn.
- We take care of our environment by sweeping the floor, picking up litter (like sweet wrappers), and throwing dirt in the dustbin. This keeps us healthy.

EVALUATION

1. Where do we cook food in the house?
2. Where do we learn in school?
3. Who cooks food at home?
4. Where should we throw dirt?
5. Mention one person you see at school.

ASSIGNMENT

1. Draw a broom used for sweeping.
2. Write the name of your best friend.
3. Ask your parents: What is the name of our neighbour?



WEEK 3: EXPLORING OTHER PARTS OF THE SURROUNDING: ROADS

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify a road as a place where cars and people move.
2. Mention two types of roads (tarred and untarred).
3. State the meaning of the three colours of the traffic light.
4. Mention one safety rule when crossing the road.

CONTENT

1. Identifying Roads

- A road is a wide path connecting one place to another, like from home to school or from one town to another.
- Cars, buses, lorries (big trucks), motorcycles (okada), bicycles, and people use the road to move.



2. Types of Roads

- **Tarred Road:** These are black, smooth, and hard because they are made with tar (asphalt). Cars move fast and smoothly on them. Most big roads in cities are tarred. Example: The main road in front of our school.
- **Untarred Road:** These are made of sand, stones, or earth. They can be dusty in dry weather and muddy when it rains. Cars move slower and can bump. Example: A path in the village.
- **Footpath:** A small, narrow path for walking only. Bicycles and people can use it, but not cars.

3. Road Safety and Traffic Lights

- We must be very careful on the road so we do not get hurt by vehicles.
- Before crossing any road, we must:
 - a. **Stop** at the edge.
 - b. **Look Left** for cars.
 - c. **Look Right** for cars.
 - d. **Look Left Again.**
 - e. If no car is coming, we walk quickly across. Do not run!
- **Traffic Light Colours:** Traffic lights tell cars and people when to stop and go.
 - **Red Light: STOP.** Cars must stop. Pedestrians (people walking) must not cross.
 - **Yellow (Amber) Light: GET READY.** It means the light is about to change. Be ready to stop or go.
 - **Green Light: GO.** Cars can move. People can cross if it is safe.

4. People Who Work on Roads

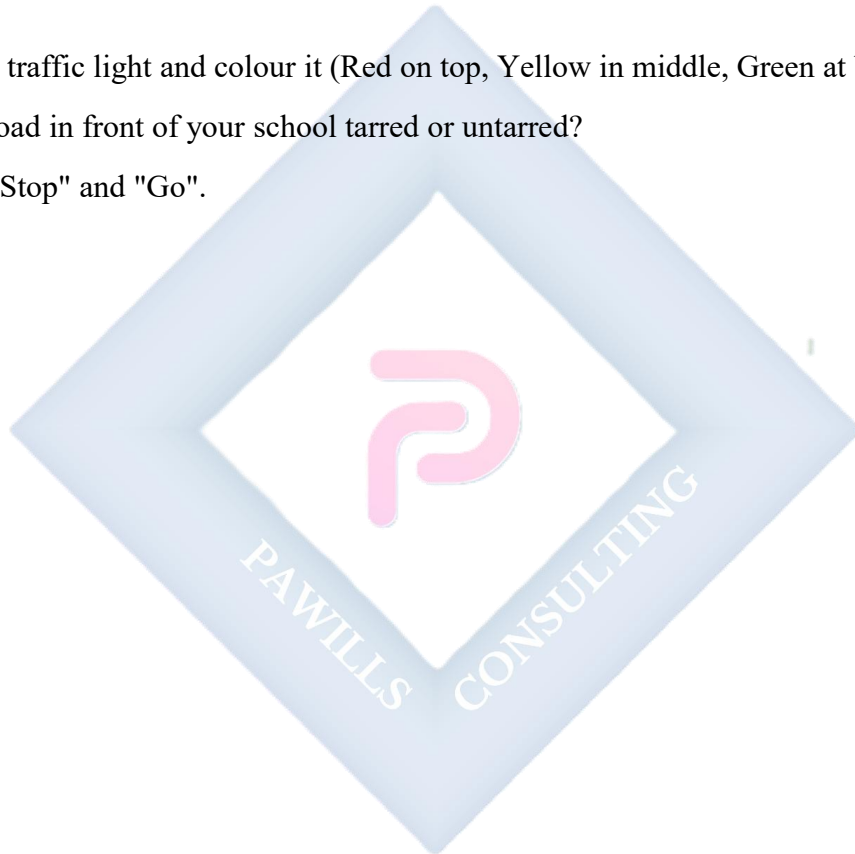
- **Drivers:** They drive the cars, buses, and lorries.
- **Traffic Wardens/Police:** They direct the cars to move or stop, especially when the traffic light is not working. They wear uniforms.
- **Construction Workers:** They fix potholes (holes in the road), build new roads, and sweep the roads. They wear helmets for safety.

EVALUATION

1. What travels on the road?
2. What colour of the traffic light says "Stop"?
3. What does the Green light mean?
4. What should you do before crossing the road?
5. Name one person who works on the road.

ASSIGNMENT

1. Draw a traffic light and colour it (Red on top, Yellow in middle, Green at bottom).
2. Is the road in front of your school tarred or untarred?
3. Write "Stop" and "Go".



WEEK 4: COLOUR IDENTIFICATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify and name the three primary colours.
2. Identify at least three other common colours.
3. Mention the colour of common objects (sky, grass, sun).
4. Group objects according to their colours.

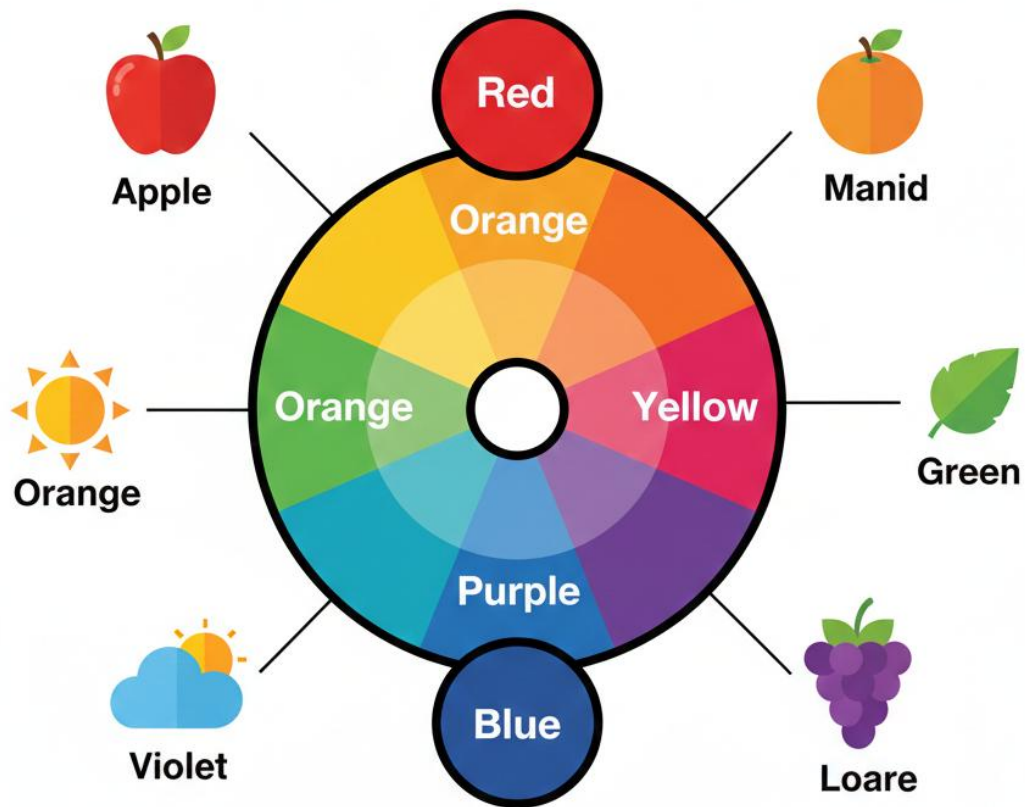
CONTENT

1. Primary Colours

- These are the main or mother colours. We cannot mix other colours to get them. All other colours come from these three.
- **Red:** Like an apple, a ripe tomato, blood, a red pen, or a stop sign.
- **Blue:** Like the clear sky on a sunny day, the deep ocean, a blue shirt, or a blue whale.
- **Yellow:** Like a ripe banana, the bright sun, a lemon, a canary bird, or a school bus.

2. Secondary and Other Colours

- When we mix two primary colours, we get new colours called secondary colours.
 - **Green:** Mix Blue + Yellow. Like grass, leaves, a frog, or a tree.
 - **Orange:** Mix Red + Yellow. Like an orange fruit, a carrot, or a tiger.
 - **Purple (Violet):** Mix Red + Blue. Like grapes, some flowers, or an eggplant.
- We also have other common colours:
 - **White:** Like chalk, paper, milk, teeth, clouds.
 - **Black:** Like charcoal, a tyre, our hair, the night sky.
 - **Brown:** Like soil, a tree trunk, chocolate.



3. Colours in Our Environment

- The sky is **Blue** during the day.
- The grass and leaves are **Green**.
- The sun is bright **Yellow**.
- Our teeth are **White**.
- A ripe tomato is **Red**.
- A chocolate is **Brown**.

4. Sorting and Mixing

- We can sort (group) things by their colour. For example, put all red toys in one basket and blue toys in another basket.
- **Magic Colour Mixing (using paint):**
 - Red paint + Yellow paint = Orange paint.
 - Blue paint + Yellow paint = Green paint.
 - Red paint + Blue paint = Purple paint.

EVALUATION

1. What colour is a banana?
2. What colour is the sky?
3. Mention two primary colours.
4. What colour is grass?
5. If you mix Red and Yellow, what colour do you get?

ASSIGNMENT

1. Draw an apple and colour it red.
2. Draw a leaf and colour it green.
3. Find three things in your house that are white.

WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions based on the topics covered from Weeks 1–4.

CONTENT

Assessment of Weeks 1–4

- Environment and Senses.
- Home and School surroundings.
- Road safety.
- Colours.

EVALUATION

- Administer the Mid-Term Test.

ASSIGNMENT

- Nil.

WEEK 6: MID-TERM BREAK

OBJECTIVES

By the end of the week, pupils should be able to:

1. Rest and refresh for the second half of the term.

CONTENT

Holiday/Rest

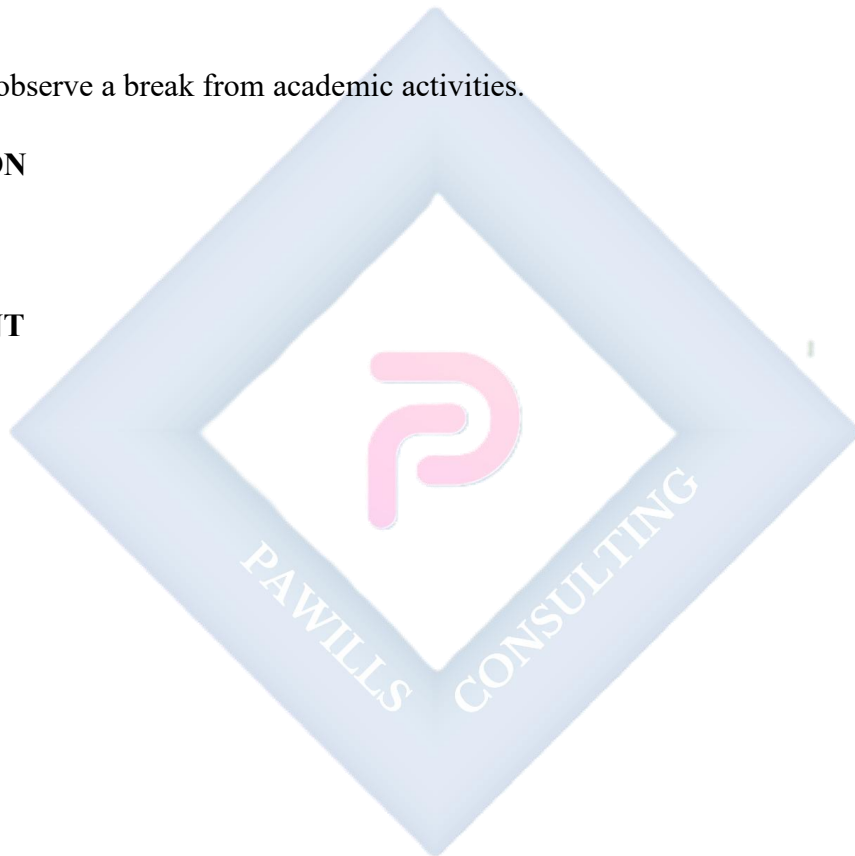
- Pupils observe a break from academic activities.

EVALUATION

- Nil.

ASSIGNMENT

- Nil.



WEEK 7: SHAPES (PART 1)

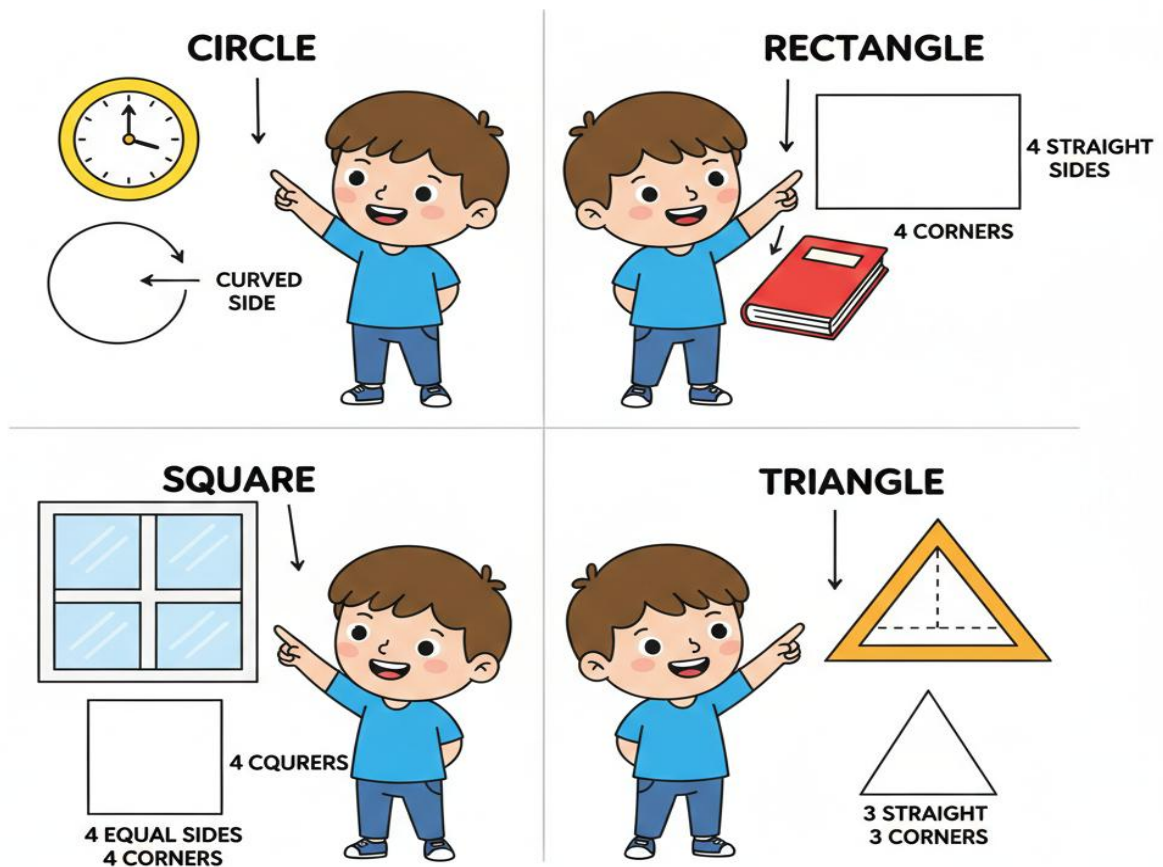
OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify and name basic shapes (Circle, Square, Triangle, Rectangle).
2. Describe a shape by counting its sides.
3. Find examples of shapes in the classroom.
4. Trace and draw basic shapes.

CONTENT

LEARNING SHAPES IN THE CLASSROOM



1. Identifying Basic Shapes

- **Circle:** It is perfectly round like a ball, a coin, or a plate. It has **no corners (edges)** and **no straight sides**. It has one curved line all around.
- **Square:** It looks like a box or a tile. It has **4 equal sides** (all sides are the same length) and **4 corners (right angles)**.
- **Triangle:** It has **3 straight sides** and **3 corners**. It looks like a slice of pizza, a party hat, or a mountain.
- **Rectangle:** It looks like a door or a book. It has **4 sides**. The opposite sides are equal (two long sides and two short sides). It has **4 corners**.

2. Properties of Shapes

- A **Circle** goes round and round. You can roll it.
- A **Square** has 4 corners and 4 equal sides. All sides are the same.
- A **Triangle** has 3 corners and 3 sides. It is pointed.
- A **Rectangle** has 4 corners. Its sides are not all equal; two are long, two are short.

3. Finding Shapes Around Us

- **Circle:** Clock face, Button, Plate, Wheel, Orange.
- **Rectangle:** Door, Blackboard, Book, Table top, Window.
- **Square:** Floor tile, Handkerchief, Picture frame, Chocolate bar.
- **Triangle:** Samosa, Roof of a house, Traffic sign, A slice of watermelon.

EVALUATION

1. Which shape is round like a ball?
2. How many sides does a triangle have?
3. Does a square have 3 sides or 4 sides?
4. What shape is the blackboard?
5. Draw a circle in the air with your finger.

ASSIGNMENT

1. Draw a Triangle and color it Blue.
2. Draw a Square and color it Red.
3. Look at the door in your house. What shape is it?



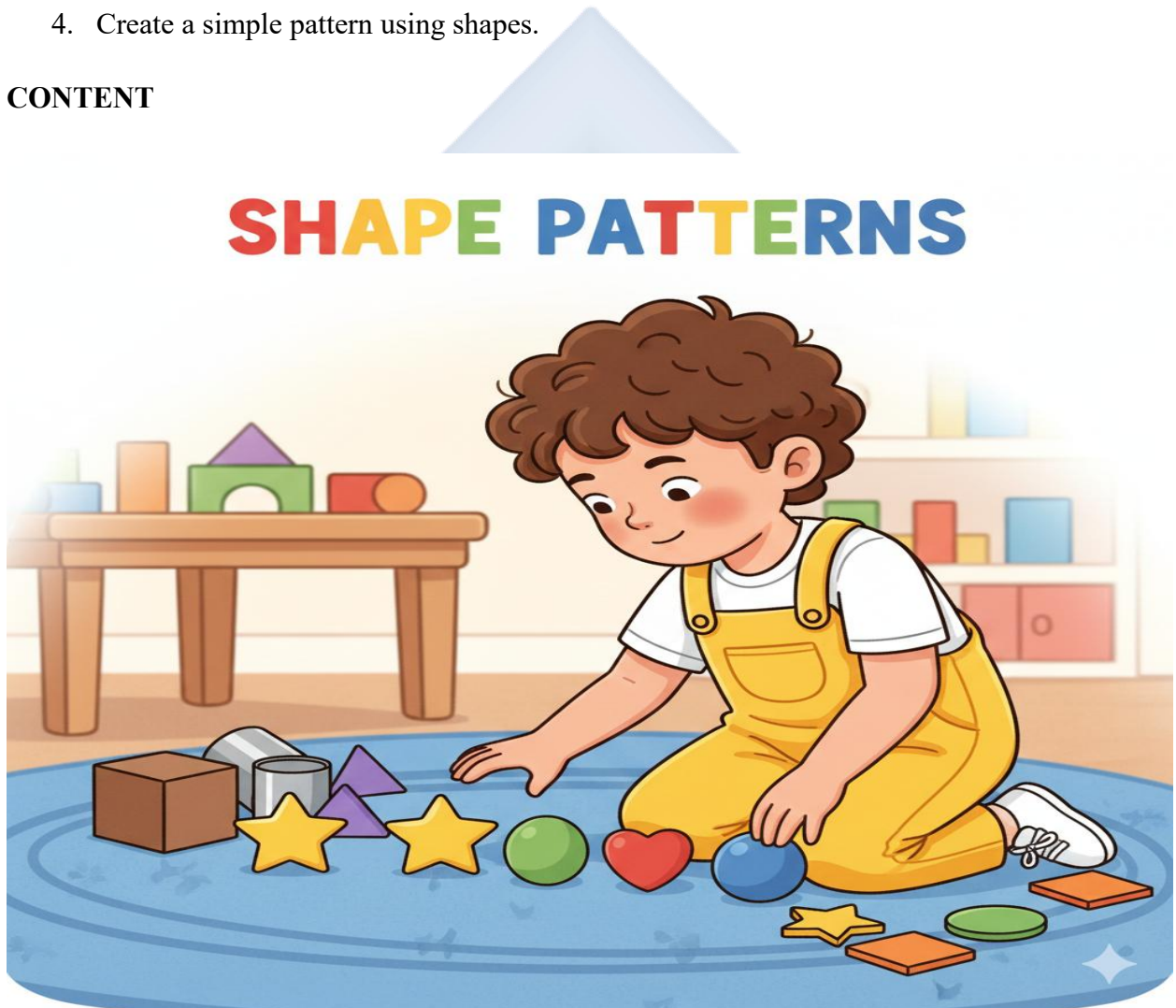
WEEK 8: SHAPES (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify special shapes (Star, Heart, Oval).
2. Differentiate between 2D (flat) and 3D (solid) shapes.
3. Identify a sphere, cube, and cylinder.
4. Create a simple pattern using shapes.

CONTENT



1. Identifying More Shapes

- **Star:** It twinkles in the sky at night. It has **5 points**. We see stars on flags and in drawings.

- **Heart:** It is the shape of love and care. It looks like two rounded parts at the top and a point at the bottom.
- **Oval:** It looks like an egg or a human face. It is a **stretched circle** or a flat, long circle.

2. Patterns with Shapes

- A pattern is when shapes or colours repeat in a certain order. It makes things look beautiful and organized.
- Example: Circle - Square - Circle - Square... (What comes next? **Circle**).
- Another example: Red, Blue, Red, Blue, Red, ____? (Blue).

3. Three-Dimensional (3D) Shapes

- **2D Shapes (Flat):** These shapes are flat like a drawing on paper. They have length and width but no thickness. Examples: Circle, Square (drawn on paper).
- **3D Shapes (Solid):** These shapes are not flat; you can hold them. They have length, width, and **height/thickness**. They are like real objects.
 - **Sphere:** A ball, an orange, a globe. It is round all over and can roll in any direction.
 - **Cube:** A box, a die (ludo dice), a sugar cube. It has **6 flat square faces**, **8 corners**, and **12 edges**. All sides are equal.
 - **Cylinder:** A tin of milk, a coke can, a drum. It has **two flat circular faces** (top and bottom) and **one curved surface**. It can roll.

EVALUATION

1. What shape looks like an egg?
2. What shape is a football? (Sphere or Circle?)
3. Draw a Star.
4. What shape is a milk tin?
5. Complete the pattern: Triangle, Star, Triangle, Star, _____.

ASSIGNMENT

1. Draw a Heart shape.

2. Bring an empty matchbox or small carton to school (This is a cuboid/cube).
3. Find something in your house that is shaped like a Cylinder.



WEEK 9: LIVING THINGS (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

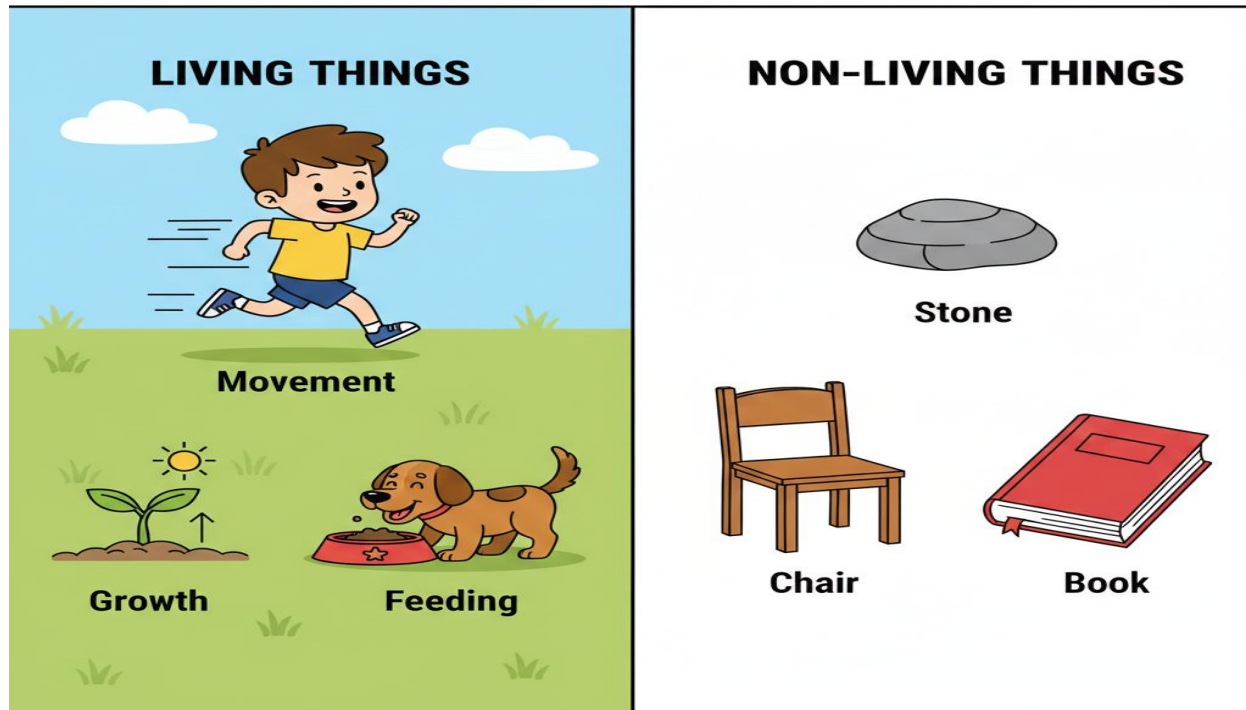
1. Define living things as things that have life.
2. Mention three characteristics of living things.
3. List three examples of living things.
4. Differentiate between living and non-living things.

CONTENT

1. What are Living Things?

- Living things are things that are **alive**. They have life in them. They are not made by people; they are created by nature/God.

LIVING vs. NON-LIVING



2. Characteristics of Living Things All living things (humans, animals, plants):

- **Movement:** They can move from one place to another (like a boy running) or move parts of their body (like a plant turning towards the sun).
- **Respiration (Breathing):** They breathe air. Humans and animals use noses/lungs. Plants breathe through tiny holes in their leaves.
- **Nutrition (Feeding):** They eat food and drink water to get energy and grow. Plants make their own food using sunlight.
- **Growth:** They grow from small to big. A baby grows into an adult. A seed grows into a big tree.
- **Reproduction:** They have babies or young ones. A cat gives birth to kittens. A plant produces seeds.
- **Sensitivity:** They can feel and respond to things. You feel pain if you touch something hot. A plant's roots grow towards water.
- **Death:** All living things die eventually when their life ends.

3. Examples of Living Things

- **Human beings:** Boys, Girls, Men, Women, Teachers.
- **Animals:** Dogs, Cats, Birds, Fish, Goats, Butterflies.
- **Plants:** Mango tree, Grass, Flowers, Maize plant, Hibiscus.

4. Living vs. Non-Living

- **Living Thing (Example: A Goat):** It moves, eats grass, breathes, grows, and has kids.
- **Non-Living Thing (Example: A Stone):** It cannot move by itself, does not eat, does not breathe, does not grow, and cannot have babies.
- **Other Non-Living Examples:** Chair, Book, Car, Water.

EVALUATION

1. What do living things eat?
2. Can a stone breathe?
3. Mention two living things.
4. Do living things grow?

5. Is a table a living thing? Why?

ASSIGNMENT

1. Draw a living thing (e.g., a tree or a fish).
2. Draw a non-living thing (e.g., a cup).
3. Write "Yes" or "No": Do cars breathe?



WEEK 10: LIVING THINGS (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Group animals into pets, farm animals, and wild animals.
2. Identify the basic parts of a plant.
3. List three things living things need to survive.
4. State one way to care for plants or animals.

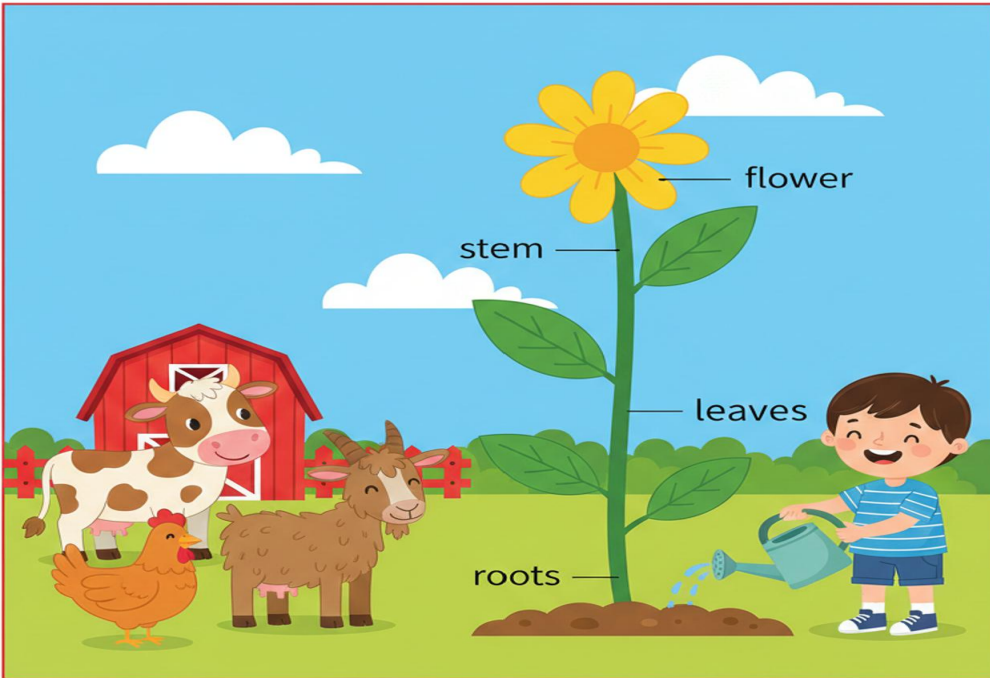
CONTENT

1. Animals as Living Things

- **Pets:** Animals we keep at home as friends and companions. They are usually tame and friendly. Examples: Dog, Cat, Rabbit, Parrot, Goldfish. We love and play with them.
- **Farm Animals (Domestic Animals):** Animals we keep on a farm. They give us food (like milk, eggs, meat) or help with work. Examples: Cow (milk), Goat (meat), Chicken (eggs), Sheep (wool), Donkey (carries load).
- **Wild Animals:** Animals that live freely in the bush, forest, jungle, or water. They find their own food and can be dangerous. Examples: Lion, Elephant, Snake, Monkey, Crocodile, Giraffe. We see them in zoos or on TV.

2. Plants as Living Things

- Plants are living things because they grow, need water, and can reproduce (make seeds).
- **Basic Parts of a Plant:**



- **Roots:** Under the ground. They hold the plant firmly and **drink water and minerals** from the soil.
- **Stem:** Stands up straight. It holds the plant up and carries water and food between the roots and leaves. Some stems are soft (herbs), some are hard (tree trunk).
- **Leaves:** Usually green parts. They **catch sunlight** to help the plant make its own food (photosynthesis). They also breathe.
- **Flowers:** The beautiful, colorful part. They attract insects and later turn into fruits and seeds.

3. What Living Things Need All living things need these to stay alive and healthy:

- **Food:** To get energy to move, grow, and stay alive. Humans eat rice, animals eat grass, plants make food from sunlight.
- **Water:** To drink, to help their bodies work, and to cool down. Plants wilt without water.
- **Air (Oxygen):** To breathe. We breathe in oxygen from the air.
- **Shelter:** A safe place to live and rest. Humans need houses, birds need nests, rabbits need burrows.

4. Care for Living Things

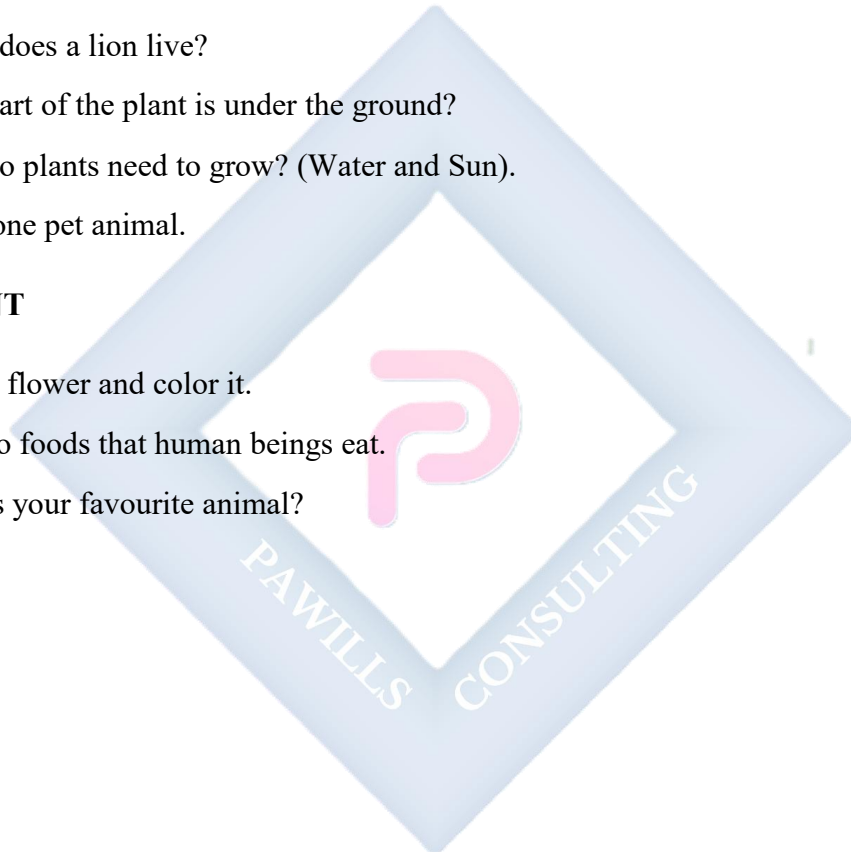
- **For Plants:** Water them regularly, put them where there is sunlight, and don't pluck their leaves unnecessarily.
- **For Animals (Pets):** Give them food and clean water, keep their living space clean, play with them gently, and take them to the animal doctor if sick.
- We should be kind to all living things.

EVALUATION

1. Name one animal that lives on a farm.
2. Where does a lion live?
3. What part of the plant is under the ground?
4. What do plants need to grow? (Water and Sun).
5. Name one pet animal.

ASSIGNMENT

1. Draw a flower and color it.
2. List two foods that human beings eat.
3. What is your favourite animal?



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts taught throughout the term.
2. Demonstrate readiness for the end-of-term examination.

CONTENT



Review of All Topics

- **Environment:** Meaning, using five senses (Sight, Hearing, Smell, Taste, Touch), things at home and school.
- **People:** Family, school people, community people.
- **Roads:** Types (tarred, untarred), safety rules (look left-right-left), traffic lights (Red=Stop, Yellow=Ready, Green=Go).
- **Colours:** Primary (Red, Blue, Yellow), others (Green, Orange), mixing colours.
- **Shapes:** 2D shapes (Circle, Square, Triangle, Rectangle, Star, Heart, Oval), 3D shapes (Sphere, Cube, Cylinder).

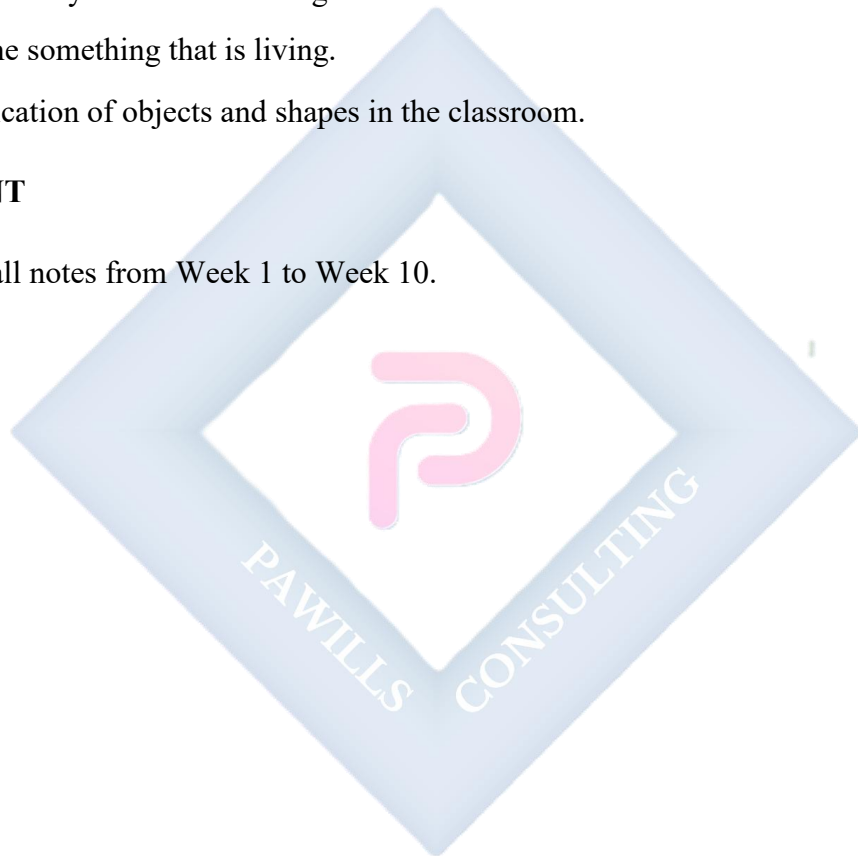
- **Living Things:** Characteristics (Move, Breathe, Eat, Grow, Reproduce), examples (Humans, Animals, Plants), needs (Food, Water, Air, Shelter).

EVALUATION

- Oral quiz covering all topics. Example questions:
 - What do you use to smell a flower?
 - What colour do you get if you mix blue and yellow?
 - How many sides does a triangle have?
 - Name something that is living.
- Identification of objects and shapes in the classroom.

ASSIGNMENT

- Study all notes from Week 1 to Week 10.



PRIMARY 1 BASIC SCIENCE SCHEME OF WORK

SECOND TERM

Week	Topic	Breakdown (Subtopics)
1	Non-Living Things (Part 1)	• Understanding what non-living things are • Characteristics of non-living things: they do not move on their own, do not eat, do not grow, do not breathe • Examples of non-living things: stones, water, air, chairs, tables, books • Identifying non-living things in our environment • Differences between living and non-living things
2	Non-Living Things (Part 2)	• Natural non-living things: rocks, water, soil, air, sand, mountains • Man-made non-living things: cars, houses, chairs, tables, toys, clothes • Uses of non-living things in our daily life • Sorting and classifying non-living things • How non-living things help living things (water for drinking, air for breathing, shelter from houses)
3	Exploring Other Parts of the Surroundings: Air	• Understanding what air is • Air is all around us (we cannot see it but we can feel it) • Feeling air: blowing, waving, fanning • Uses of air: breathing, drying clothes, flying kites, inflating balloons • Moving air is called wind • Clean air and dirty air (pollution –

		introduction)
4	Exploring Other Parts of the Surroundings: Soil	• Understanding what soil is • Where we find soil: ground, gardens, farms, pots • Types of soil: sandy soil, clay soil, loamy soil (basic observation) • Colours of soil: black, brown, red, yellow • Uses of soil: growing plants, making pots, building houses (mud) • Taking care of soil (not littering, planting trees)
5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday/Rest
7	Exploring Other Parts of the Surroundings: Water	• Understanding what water is • Sources of water: rain, river, stream, well, borehole, tap • Forms of water: liquid (drinking water), solid (ice), gas (steam from boiling water) • Uses of water: drinking, cooking, bathing, washing, cleaning, watering plants • Clean water and dirty water
8	Water (Continued) & Water Safety	• Importance of clean water for health • Ways to keep water clean: covering water containers, not throwing dirt in water • Water safety: not playing near deep water, not drinking dirty water • Saving water: turning off taps, not wasting water • Water cycle (very simple introduction): rain falls, water flows to rivers, sun dries water, rain falls again

9	Energy (Part 1)	• Understanding what energy is (the ability to do work or make things happen) • Energy makes things work, move, or change • Sources of energy: sun, food, batteries, electricity • Energy from the sun: gives light, gives heat, helps plants grow, dries clothes • Energy from food: helps us run, jump, play, think
10	Energy (Part 2)	• Energy from batteries: makes toys work, torches give light, clocks work • Energy from electricity: lights our homes, powers television, charges phones, runs fans • Safety with electricity: not touching sockets, not playing with wires, telling adults about electrical problems • Saving energy: switching off lights when not in use, not wasting food • Renewable energy (simple awareness): sun, wind, water
11	REVISION	• Review of all topics: Non-living things, air, soil, water, energy
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends

PRIMARY 1 BASIC SCIENCE LESSON NOTES

SECOND TERM

WEEK 1: NON-LIVING THINGS (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define non-living things in simple terms.
2. Mention three characteristics of non-living things.
3. List four examples of non-living things found in the classroom or home.
4. State one difference between living and non-living things.

CONTENT

1. Understanding Non-Living Things

- Non-living things are things that **do not have life** in them. They are **lifeless**.
- They cannot do the things that humans, animals, and plants can do. They are not born and they do not die like living things.

Living vs. Non-Living Things

LIVING THINGS	NON-LIVING
 Eat Grow Move	 Don't Eat Don't Grow
LIVING	NON-LIVING
• Have life • Move	• Don't Move • Eat • Eat

2. Characteristics of Non-Living Things

- **Movement:** They **cannot move on their own**. A stone will stay in one place unless you kick it, throw it, or the wind blows it.
- **Feeding:** They **do not eat food or drink water**. A table does not get hungry or thirsty.
- **Growth:** They **do not grow**. A small chair will not become a big chair over time.
- **Breathing:** They **do not breathe air**. They do not have noses, lungs, or need oxygen.
- **Reproduction:** They **do not have babies or young ones**. A pencil cannot give birth to another pencil.

3. Examples of Non-Living Things

- **In the classroom:** Tables, chairs, books, pencils, blackboard, chalk, duster, bag, ruler.
- **In the home:** Television, radio, cup, plate, spoon, bed, fan, clock, door.
- **In the environment:** Stones, sand, water, air, mountains, cars, bicycles, bottles, plastic.

4. Differences between Living and Non-Living Things

Living Things	Non-Living Things
Have life	Do not have life
Can move by themselves	Cannot move by themselves
Eat food and drink water	Do not eat or drink
Grow (from small to big)	Do not grow
Breathe	Do not breathe
Reproduce (have young ones)	Do not reproduce

EVALUATION

1. What are non-living things?
2. Can a stone move by itself?
3. Do non-living things eat food?
4. Mention three non-living things in your bag.

5. Does a pencil breathe?
6. Mention one difference between a dog (living) and a toy dog (non-living).

ASSIGNMENT

1. Draw a stone and a cup in your notebook.
2. List three non-living things you can see in your kitchen.
3. Complete this sentence: Non-living things do not _____. (eat / move / grow).



WEEK 2: NON-LIVING THINGS (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Differentiate between natural and man-made non-living things.
2. List three examples of natural non-living things.
3. List three examples of man-made non-living things.
4. State two ways non-living things help living things.

CONTENT

1. Natural Non-Living Things

- These are things **created by nature (or God)**. People did not make them. They are found in the environment.
- **Examples:** Stones, rocks, sand, soil, water (in rivers), air, mountains, seashells, gold, sunlight, rainwater.



2. Man-Made Non-Living Things

- These are things **made by human beings** (people) to help us live better, work easier, or have fun.
- **Examples:** Cars, houses, chairs, tables, pencils, toys, clothes, shoes, fans, television, books, spoons.

3. Sorting Non-Living Things

- We can group things by asking: "**Did a person make this?**"
 - If **YES**, it is **man-made**. Example: A plastic bottle.
 - If **NO**, it is **natural**. Example: A piece of wood from a tree (before it is cut and shaped).
- Some things start as natural but become man-made. Example: Wood (natural) is used to make a chair (man-made).

4. Uses of Non-Living Things

Non-living things are very important for living things:

- **Water:** We drink it to stay alive, use it for cooking, bathing, and washing.
- **Air:** We breathe it to stay alive. All animals and plants need air.
- **Houses (Shelter):** They protect us from rain, sun, cold, and dangerous animals.
- **Clothes:** They cover our bodies, keep us warm, and make us look nice.
- **Cars and Bicycles:** They help us move from one place to another quickly and without getting tired.
- **Spoons and Plates:** They help us eat our food neatly.

EVALUATION

1. Is a stone man-made or natural?
2. Is a car man-made or natural?
3. Mention two non-living things that are natural.
4. Mention two non-living things that are man-made.
5. Why do we need houses?

6. Can we live without air and water?

ASSIGNMENT

1. Draw one natural non-living thing (e.g., a rock).
2. Draw one man-made non-living thing (e.g., a ball).
3. Write down one use of water.



WEEK 3: EXPLORING OTHER PARTS OF THE SURROUNDINGS: AIR

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define air as something we feel but cannot see.
2. Demonstrate how to feel air.
3. Define wind.
4. Mention three uses of air.

CONTENT

1. Understanding Air

- Air is **everywhere around us**. It is in the classroom, outside, in our homes, and even in empty-looking spaces.
- We **cannot see air** with our eyes because it is colourless and transparent (clear).
- We **cannot hold air** in our hands like we hold a stone, but we can **feel it** on our skin.



2. Feeling Air (Air is Real)

- Air moves and when it moves against our skin, we feel it. This shows air is real even though we can't see it.
 - **Fanning:** Fan yourself with a book or your hand. You feel cool air on your face.
 - **Blowing:** Blow on the palm of your hand. You feel the air coming from your mouth.
 - **Seeing its Effect:** Watch leaves shaking on a tree, a flag fluttering, or papers flying. The moving air (wind) is pushing them.

3. Wind

- When air **moves fast**, we call it **Wind**.
- Wind can be gentle (a breeze) or strong (a storm).
- Wind can do many things: blow papers away, make kites fly, turn windmill blades, and cool us on a hot day.

4. Uses of Air

- **Breathing:** The most important use. All living things (people, animals, plants) need air (oxygen) to live.
- **Drying:** Air helps to dry our wet clothes, towels, and hair after washing.
- **Flying:** Air helps birds, insects, aeroplanes, and kites to fly.
- **Filling Things:** We pump air into balloons to make them big, into footballs to make them bouncy, and into car/bicycle tyres to make them round and strong.
- **Moving Things (Wind Energy):** Wind can push sailboats, turn windmills to generate electricity, and scatter plant seeds.

5. Clean and Dirty Air

- **Good (Clean) Air:** Is fresh and has no bad smell. It is good for our health. We find it in gardens and places with many trees.
- **Bad (Dirty/Polluted) Air:** Has smoke (from cars, factories, burning), dust, and bad smells. Breathing dirty air can make us cough and sick. We should avoid burning rubbish and plant more trees for clean air.

EVALUATION

1. Can you see air?
2. How can you feel air?
3. What is moving air called?
4. Mention one thing we fill with air.
5. Why do we hang wet clothes outside?
6. Is smoke good for the air?

ASSIGNMENT

1. Use a piece of paper to fan yourself. Write down what you felt.
2. Draw a balloon filled with air.
3. List two things that fly in the air.



WEEK 4: EXPLORING OTHER PARTS OF THE SURROUNDINGS: SOIL

OBJECTIVES

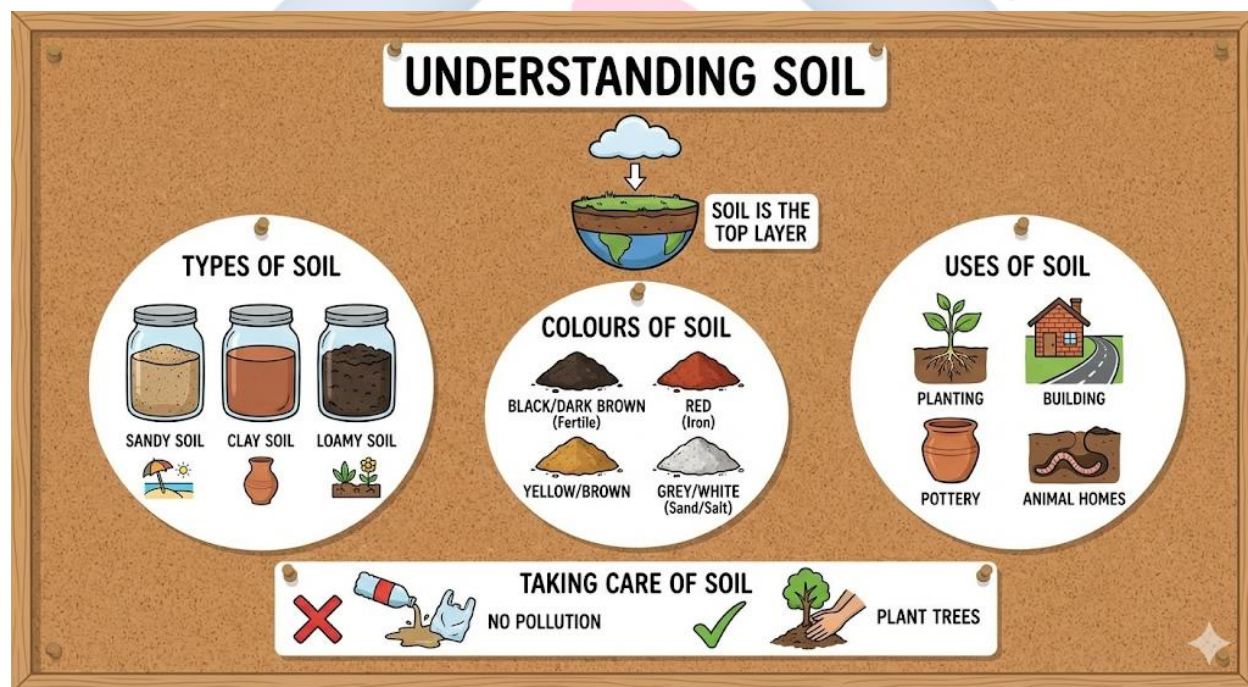
By the end of the lesson, pupils should be able to:

1. Define soil as the ground we walk on.
2. Identify three types of soil based on simple observation.
3. Mention three colours of soil.
4. List three uses of soil.

CONTENT

1. Understanding Soil

- Soil is the **top, soft layer of the earth**. It is the ground where we stand, walk, and play.
- We find soil in the garden, on the farm, in flower pots, and under the grass.



2. Types of Soil

- **Sandy Soil:**
 - **Feel:** Rough, gritty, and loose. It does not stick together.

- **Where found:** Beaches, sandy areas.
- **Property:** Water passes through it very quickly. Not very good for holding water for plants.
- **Clay Soil:**
 - **Feel:** Smooth, sticky, and heavy when wet. It can be moulded into shapes like pottery.
 - **Where found:** In some riverbanks and valleys.
 - **Property:** Holds water very well, sometimes too well, so it can become waterlogged.
- **Loamy Soil:**
 - **Feel:** Soft, crumbly, and dark. It is a mixture of sand, clay, and decayed plants/animals (humus).
 - **Where found:** Good gardens and farms.
 - **Property:** **The best soil for planting** because it holds water and nutrients well but also lets air reach plant roots.

3. Colours of Soil

- Soil can be different colours depending on the minerals and organic matter in it.
 - **Black/Dark Brown:** Very fertile (good for plants). Has lots of humus.
 - **Red:** Contains iron.
 - **Yellow/Brown:** Common colour for many gardens.
 - **Grey/White:** May have a lot of sand or salt.

4. Uses of Soil

- **Planting (Agriculture):** Plants grow in the soil. Soil gives them water, nutrients, and support.
- **Building:** We use soil (especially sand and clay) mixed with other materials to build houses (mud houses, bricks), roads, and dams.
- **Pottery:** Clay soil is used by potters to make beautiful pots, plates, cups, and vases.
- **Home for Animals:** Many small animals live in the soil, like earthworms, ants, beetles, and rabbits (in burrows). They help make the soil healthy.
- **Making Things:** Children use sand to build sandcastles.

5. Taking Care of Soil

- We should not pour dirty oil, chemicals, or throw plastic rubbish on the soil. It kills the tiny living things in it and makes it bad for plants.
- We should plant trees and grasses. Their roots hold the soil together and stop it from being washed away by rain (erosion).

EVALUATION

1. Where do we find soil?
2. Which soil is sticky and used for moulding?
3. Which soil is best for planting?
4. Mention two colours of soil.
5. Mention one animal that lives in the soil.
6. What do we use to build houses?

ASSIGNMENT

1. Collect a small sample of soil from near your house and bring it to school (in a small nylon/bag).
2. Draw a flower pot with soil and a plant growing in it.
3. Write down one use of soil.

WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions based on the topics covered from Week 1 to Week 4.

CONTENT

Assessment of Weeks 1–4

- Non-Living Things (Characteristics and Examples).
- Natural vs. Man-made things.
- Air (Wind and Uses).
- Soil (Types and Uses).

EVALUATION

- Administer the Mid-Term Test.

ASSIGNMENT

- Nil.

WEEK 6: MID-TERM BREAK

OBJECTIVES

By the end of the week, pupils should be able to:

1. Rest and refresh for the second half of the term.

CONTENT

Holiday/Rest

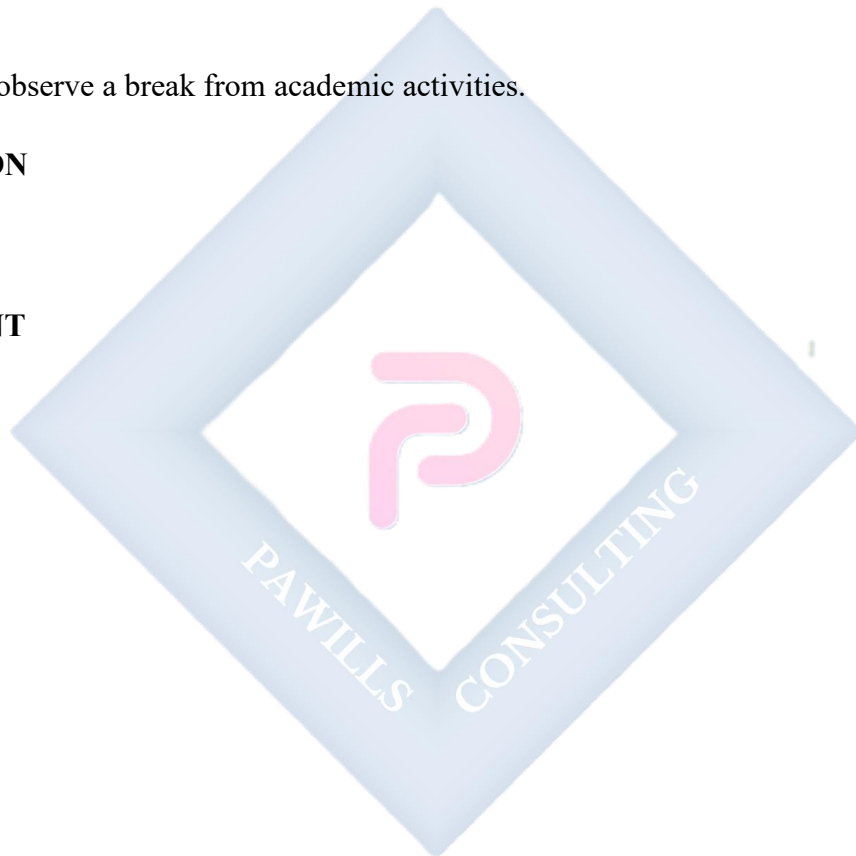
- Pupils observe a break from academic activities.

EVALUATION

- Nil.

ASSIGNMENT

- Nil.



WEEK 7: EXPLORING OTHER PARTS OF THE SURROUNDINGS: WATER

OBJECTIVES

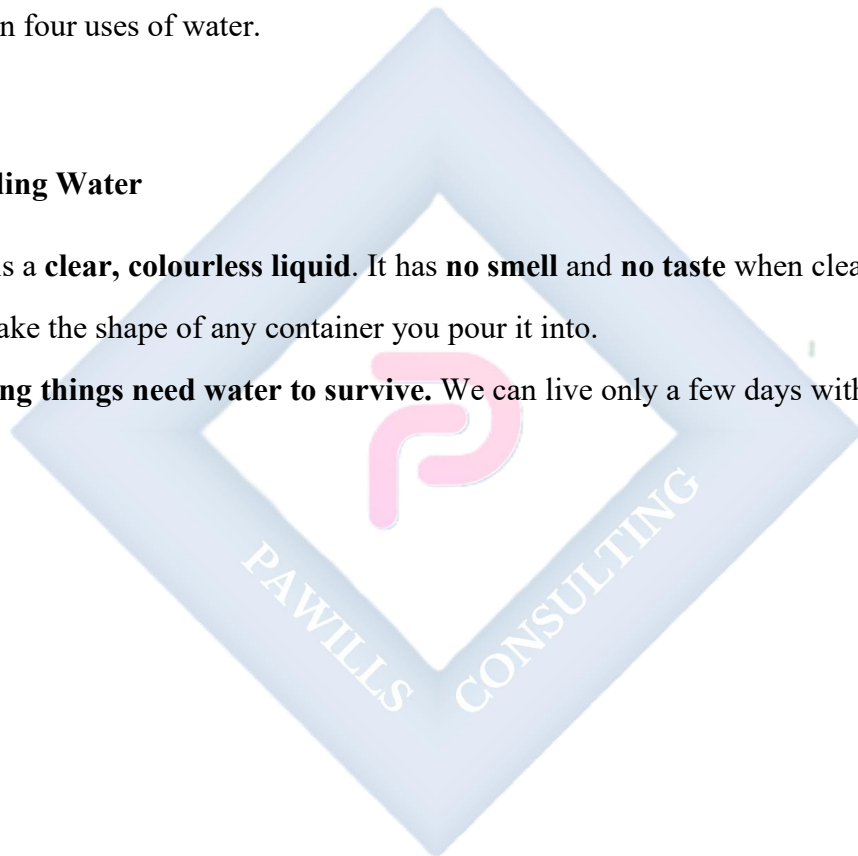
By the end of the lesson, pupils should be able to:

1. Define water as a liquid we drink.
2. List three sources of water.
3. Identify the three forms of water (liquid, solid, gas).
4. Mention four uses of water.

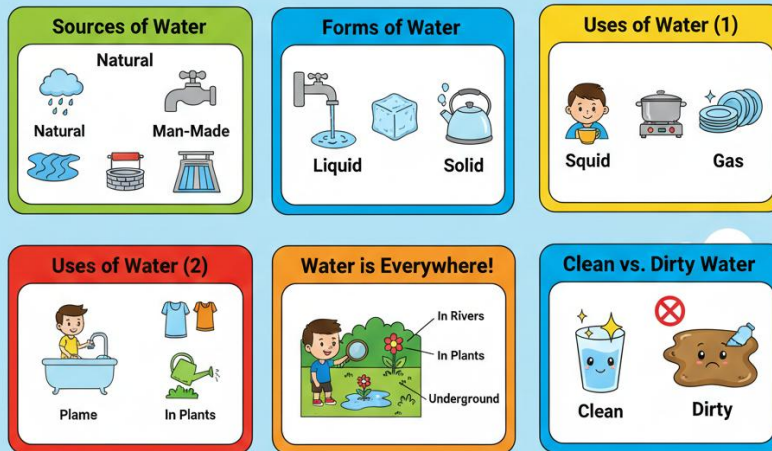
CONTENT

1. Understanding Water

- Water is a **clear, colourless liquid**. It has **no smell** and **no taste** when clean.
- It can take the shape of any container you pour it into.
- **All living things need water to survive.** We can live only a few days without water.



All About Water!



2. Sources of Water These are places where we get water from.

- **Natural Sources (Made by Nature):**
 - **Rain:** Water that falls from clouds.
 - **River:** A large natural stream of flowing water.
 - **Stream:** A small, narrow river.
 - **Spring:** Water that flows naturally from the ground.
 - **Lake/Pond:** A large/small area of water surrounded by land.
 - **Ocean/Sea:** A very large area of salty water.
- **Man-Made Sources (Made by People):**
 - **Tap:** Connected to pipes that bring water from a treatment plant or borehole.
 - **Well:** A deep hole dug in the ground to reach underground water.
 - **Borehole:** A deep, narrow hole drilled to pump water from underground.
 - **Dam:** A wall built across a river to store a large amount of water.

3. Forms of Water (States of Water)

Water can exist in three different forms:

- **Liquid:** This is the **normal water** we pour, drink, and swim in. Example: Water in a cup, river water.
- **Solid:** When liquid water gets **very cold (0°C or below)**, it freezes and becomes **Ice**. Ice is hard and has a fixed shape. Example: Ice block, ice cube, snow, hail.
- **Gas:** When liquid water is **heated (to 100°C)**, it boils and turns into an invisible gas called **Water Vapour or Steam**. We can see steam as a white mist above boiling water. Example: Steam from a boiling pot, clouds.

4. Uses of Water

- **Drinking and Cooking:** To quench thirst and prepare food (rice, soup, tea).
- **Washing:** Clothes, plates, cars, and our homes.
- **Bathing and Personal Hygiene:** To keep our bodies clean and prevent sickness.
- **Watering Plants:** Farmers and gardeners use water to help crops and flowers grow.
- **Recreation:** For swimming, playing, and boating.
- **For Animals:** All animals need to drink water to live.

EVALUATION

1. What is the main source of water from the sky?
2. Mention two places we can fetch water.
3. What happens to water when it gets very cold?
4. Mention three things we do with water at home.
5. Can we live without water?

ASSIGNMENT

1. Draw a bucket of water.
2. Ask your parents where the water in your house comes from. Write it down.
3. List two things you washed with water this week.

WEEK 8: WATER (CONTINUED) & WATER SAFETY

OBJECTIVES

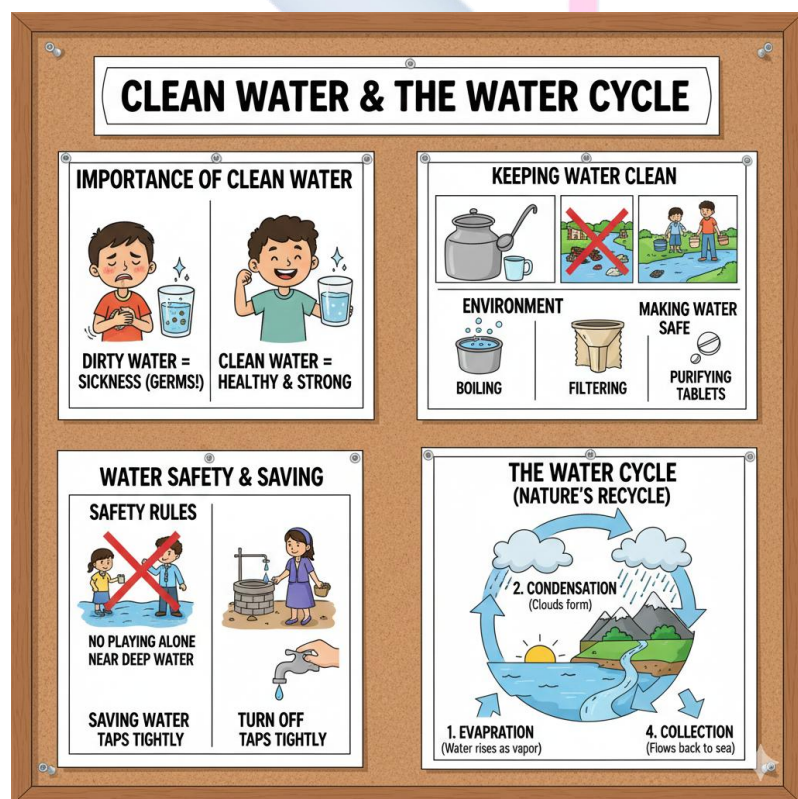
By the end of the lesson, pupils should be able to:

1. Explain why we need clean water.
2. State two ways to keep water clean.
3. Mention two safety rules regarding water.
4. Describe the water cycle simply.

CONTENT

1. Importance of Clean Water

- Dirty water contains tiny germs (bacteria and viruses) that we cannot see.
- Drinking dirty or untreated water can make us very sick with diseases like **diarrhoea, cholera, and typhoid**.
- Clean water keeps us healthy, strong, and free from stomach aches.



2. Keeping Water Clean

- **At Home:**
 - Always **cover** water stored in pots, buckets, or tanks with a clean lid to keep out dust and insects.
 - Use a clean cup or ladle to fetch water. Do not put dirty hands into the drinking water.
 - Keep the area around your water source (well, tap) clean.
- **In the Environment:**
 - Do not throw rubbish, poop, or wash clothes in rivers or streams where people fetch drinking water.
 - Do not bath or ease yourself near water sources.
- **Making Water Safe to Drink:**
 - **Boiling:** Heat water until it bubbles (boils) for some minutes. This kills germs.
 - **Filtering:** Pass water through a clean cloth or a water filter to remove dirt.
 - **Using Water Purifier Tablets:** Sometimes special tablets are added to kill germs.

3. Water Safety Rules

- **Safety Near Water:**
 - **Never play alone near deep water** like wells, rivers, ponds, or during heavy rain floods. You could fall in and drown.
 - Always go with an adult to fetch water from a well or river.
- **Safety for Drinking Water:**
 - **Do not drink water from the floor, gutters, or flowing drains.** It is very dirty.
 - Do not drink water that looks dirty, has a bad smell, or has things floating in it.
- **Saving Water:**
 - **Always turn off the tap tightly** after using it. A dripping tap wastes a lot of water.
 - Use only the amount of water you need. Do not let water run unnecessarily while washing or brushing.

4. The Water Cycle (Simple)

- This is how nature **recycles and cleans water** over and over again.
 - a. **Sun heats** water in rivers, lakes, and oceans.
 - b. Water turns into **water vapour (gas)** and rises up into the sky. This is called **evaporation**.
 - c. The vapour gets cold high up and forms tiny droplets that come together to make **Clouds**. This is called **condensation**.
 - d. When the clouds get too heavy, the water falls back to the ground as **Rain (or snow)**. This is called **precipitation**.
 - e. The rain water flows into rivers and goes back to the sea, and the cycle starts again.

EVALUATION

1. Why is dirty water bad for us?
2. How can we make water safe to drink?
3. Should we leave the tap running when we are not using it?
4. Where does rain come from?
5. What should you use to cover your water container?

ASSIGNMENT

1. Draw a cloud with rain falling down.
2. Write one rule for water safety.
3. Draw a cup with a cover on it.

WEEK 9: ENERGY (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define energy as the ability to do work.
2. Identify the Sun as the main source of natural energy.
3. Explain why we need energy from food.
4. Mention three things the Sun helps us do.

CONTENT

1. Understanding Energy

- **Energy is the strength or power to do things, to make things move, work, or change.**
- Everything that moves or works needs energy.
 - A car needs energy from petrol to move.
 - A bulb needs electrical energy to give light.
 - **You need energy to walk, run, play football, think, and even to breathe.**

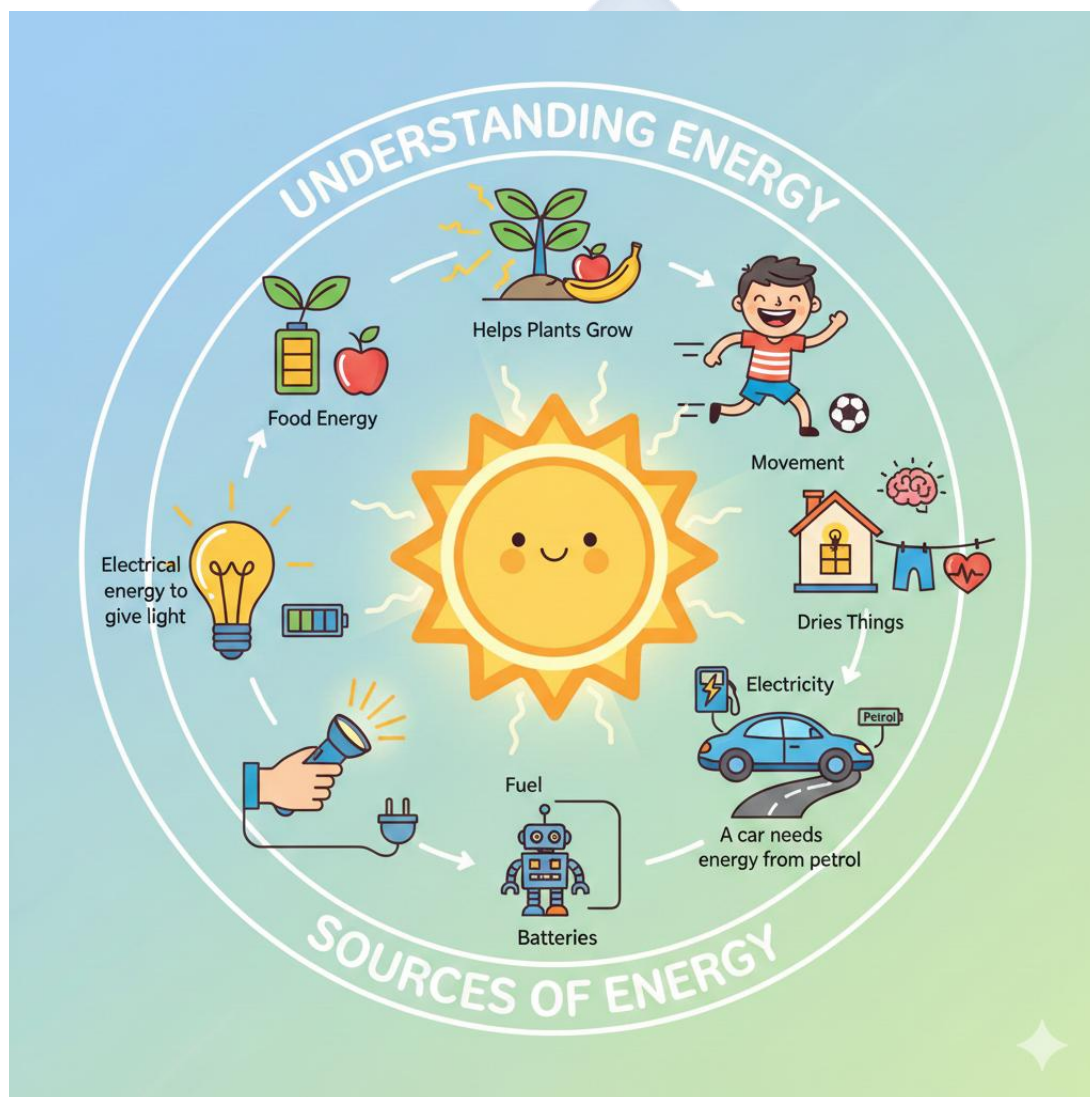
2. Sources of Energy Where we get energy from:

- **The Sun:** This is the **biggest and most important source** of natural energy for the Earth.
- **Food:** This is the source of energy for **all living things**.
- **Electricity:** A very useful form of energy that powers machines in our homes and schools.
- **Batteries:** They store chemical energy and can power small things like torches and toys.
- **Fuel:** Like petrol, diesel, kerosene. They provide energy for cars, generators, and stoves.

3. Energy from the Sun

- The Sun is like a giant ball of fire in the sky that gives us energy every day.
- **What the Sun gives us:**

- **Light Energy:** So we can see during the day. Without the sun, it would be dark all the time.
- **Heat Energy:** To keep the Earth warm. Without the sun, our world would be very, very cold.
- **Helps Plants Grow (Food Energy):** Plants use sunlight to make their own food through photosynthesis. This food gives energy to animals and humans when we eat the plants.
- **Dries Things:** The sun's heat dries our wet clothes, crops (like maize), and fish.



4. Energy from Food

- **Food is fuel for our bodies.** Just like a car needs petrol, our body needs food to work.
- When we eat foods like rice, beans, yam, bread, fruits, and vegetables, our body breaks them down to get energy.
- This energy is used for:
 - **Movement:** To run, jump, dance, and play.
 - **Growth:** To grow from a child to an adult.
 - **Body Functions:** To keep our heart beating, lungs breathing, and brain thinking.
- If you don't eat food, you will feel **tired, weak, and cannot play or learn well.**

EVALUATION

1. What is energy?
2. What is our main source of light during the day?
3. Why do we eat food?
4. Mention two things the Sun does for us.
5. What happens if you don't eat for a long time? (You get tired/weak).

ASSIGNMENT

1. Draw the Sun.
2. List your two favourite foods that give you energy.
3. Complete: The Sun gives us _____ and _____.

WEEK 10: ENERGY (PART 2)

OBJECTIVES

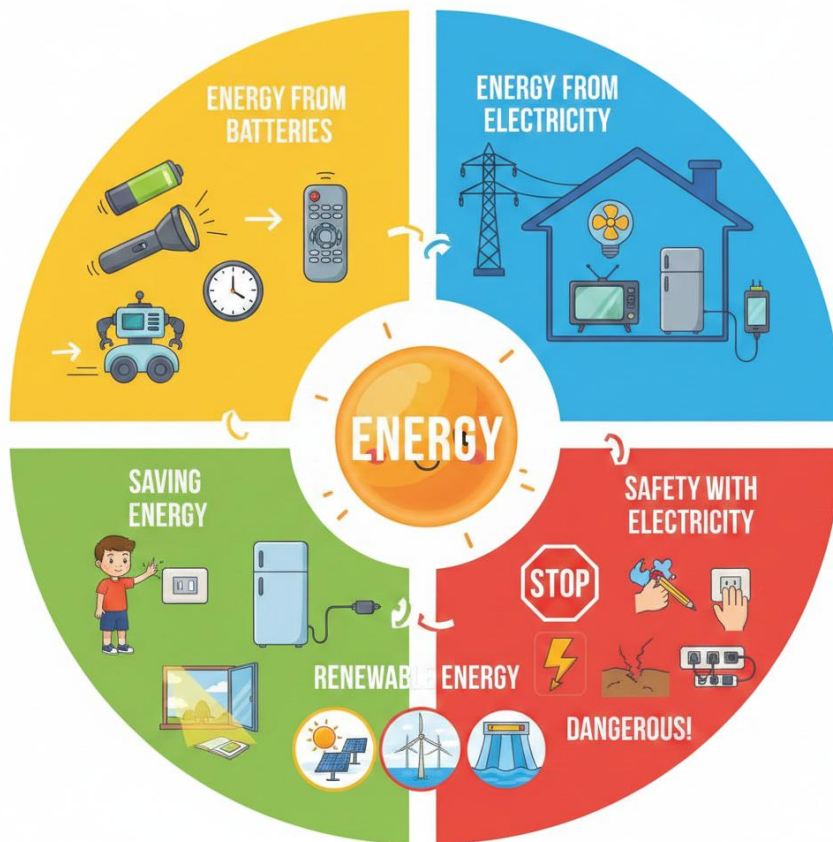
By the end of the lesson, pupils should be able to:

1. Identify objects that use batteries and electricity.
2. State two safety rules when using electricity.
3. Mention two ways to save energy.
4. List three sources of renewable energy.

CONTENT

1. Energy from Batteries

- Batteries are small containers that **store chemical energy**. When we put them in a device, they provide power.



- **Things that use batteries:**

- Torchlight (to give light)
- Wall clock (to tell time)
- TV remote control (to change channels)
- Toy cars and robots (to make them move)
- Radios and some clocks

2. Energy from Electricity

- Electricity is a powerful and very useful form of energy that flows through wires.
- It comes into our homes and schools through power lines.
- **Things that use electricity in our homes:**
 - **Lighting:** Electric bulbs, tube lights.
 - **Cooling/Air Movement:** Fans, air conditioners.
 - **Appliances:** Television, Refrigerator (fridge), Electric Iron, Blender, Microwave.
 - **Charging:** Mobile phones, tablets, laptops.
- Electricity makes our lives much easier and more comfortable.

3. Safety with Electricity

- **Electricity is DANGEROUS!** It can give a painful shock, cause severe burns, or even kill.
- **Important Safety Rules:**
 - a. **Never touch electrical sockets, plugs, or switches with WET HANDS.** Water conducts electricity and the shock will be worse.
 - b. **Never put fingers, pencils, pins, or any object into socket holes.** You could get shocked.
 - c. **Do not play with or touch broken/cut electrical wires** hanging from poles or lying on the ground. Tell an adult immediately.
 - d. Do not plug too many things into one socket.
 - e. Ask an adult for help if you need to plug or unplug something.

4. Saving Energy (Conservation)

- We should not waste energy because some sources (like fuel) can finish, and making electricity can be expensive.
- **Ways to Save Energy at Home:**
 - **Switch off lights, fans, and TVs** when you leave a room and nobody is using them.
 - **Unplug chargers** from the socket when your phone is fully charged.
 - **Don't leave the fridge door open** for a long time.
 - **Use natural light** from the sun during the day instead of turning on bulbs.

5. Renewable Energy (Simple)

- These are sources of energy that **do not finish** or get used up. They are natural and can be used again and again.
 - **Sun (Solar Energy):** Using solar panels to catch sunlight and turn it into electricity.
 - **Wind (Wind Energy):** Using big windmills (turbines) to catch wind and turn it into electricity.
 - **Water (Hydro Energy):** Using the force of flowing water in dams or rivers to turn turbines and make electricity.

EVALUATION

1. Mention two things that use batteries.
2. Mention two things that use electricity in the house.
3. Is it safe to touch a switch with wet hands?
4. How can we save energy at home?
5. Name one type of energy that does not finish (Renewable).

ASSIGNMENT

1. Draw a torchlight.
2. Check your house: List three things plugged into the wall socket.
3. Write "Safe" or "Unsafe": Playing with a cut wire.

WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from Non-living things, Air, Soil, Water, and Energy.
2. Demonstrate readiness for the end-of-term examination.

CONTENT

Review of All Topics

- **Non-Living Things:** Definition, characteristics, Natural vs Man-made examples.
- **Air:** Cannot see it, can feel it, wind, uses (breathing, drying, flying), clean vs dirty air.
- **Soil:** Types (Sandy, Clay, Loamy), colours, uses (planting, building, pottery).
- **Water:** Sources (rain, river, tap, well), forms (liquid, solid-ice, gas-steam), uses, importance of clean water, safety rules, simple water cycle.
- **Energy:** Meaning, sources (Sun, food, electricity, batteries), uses, safety with electricity, saving energy, renewable energy (Sun, Wind, Water).

EVALUATION

- Oral quiz regarding identification of objects (Is this living or non-living? Natural or man-made?).
- Simple sorting games (e.g., grouping pictures of energy sources).
- Questions on safety (What should you not do with electricity?).

ASSIGNMENT

- Study all notes from Week 1 to Week 10 in preparation for the examination.

PRIMARY 1 BASIC SCIENCE SCHEME OF WORK

THIRD TERM

Week	Topic	Breakdown (Subtopics)
1	Technology (Part 1: Understanding Technology)	• What is technology? (using tools and machines to make work easier) • Technology in our homes: television, radio, telephone, electric iron, refrigerator, fan • How technology helps us: makes work easier, saves time, makes life comfortable • Simple technology vs. complex technology • Technology for communication: telephone, mobile phone, radio, television
2	Technology (Part 2: Technology Around Us)	• Technology in schools: computers, projectors, fans, lights, bells • Technology for transportation: cars, buses, motorcycles, bicycles, trains, airplanes • Technology in hospitals: thermometer, stethoscope, X-ray machines • Technology for farming: hoes, cutlasses, tractors, watering cans • How technology has changed over time (old vs. new): old phones vs. smartphones, letter writing vs. email (basic awareness)
3	Measuring Things (Part 1: Length and Size)	• Understanding measurement (finding out how long, how big, how heavy something is) • Measuring length: long and short, tall and short • Comparing lengths: longer, shorter, longest, shortest, same length

		• Non-standard units for measuring length: hand span, foot length, pencil, stick • Measuring activities: measuring desks, books, classrooms using non-standard units
4	Measuring Things (Part 2: Weight and Capacity)	• Measuring weight: heavy and light • Comparing weights: heavier, lighter, heaviest, lightest, same weight • Using hands to feel weight differences • Introduction to simple balance scales • Measuring capacity: full, half-full, empty • Comparing capacity: holds more, holds less
5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday/Rest
7	Measuring Things (Part 3: Time and Temperature)	• Measuring time: day and night, morning, afternoon, evening • Understanding fast and slow, quick and long • Days of the week, months (awareness) • Introduction to clock: telling time (o'clock only) • Measuring temperature: hot and cold, warm and cool • Using our body to feel temperature differences
8	Simple Machines (Part 1: Introduction)	• What are simple machines? (tools that make work easier) • Why we use simple machines: to lift, move, cut, hold things • Identifying simple machines in our environment

		- Types of simple machines we will learn: lever, wheel and axle, inclined plane, pulley, wedge, screw - Simple machines at home and school
9	Simple Machines (Part 2: Types)	- Lever: seesaw, bottle opener, scissors, broom - Wheel and axle: bicycle, car wheels, door knob, wheelbarrow - Inclined plane: ramp, slope, slide, staircase - How each simple machine helps make work easier - Practical demonstrations with real objects
10	Simple Machines (Part 3: More Types and Uses)	- Wedge: knife, axe, needle, door stopper - Pulley: flag pole, well bucket, curtain rod, crane (simple observation) - Screw: bottle cap, jar lid, wood screw, light bulb - Finding simple machines in our classroom and home - How simple machines work together (compound machines – bicycle has wheels and levers) - Making simple machines: creating a simple lever with ruler and eraser
11	REVISION	Review of all topics: Technology, measuring things, simple machines
12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends

PRIMARY 1 BASIC SCIENCE LESSON NOTES

THIRD TERM

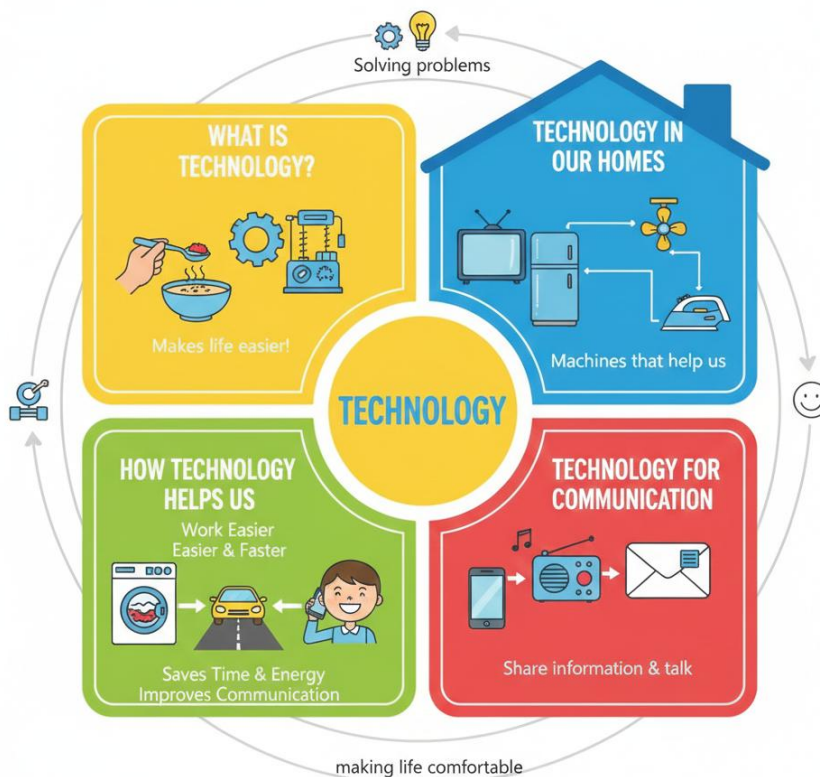
WEEK 1: TECHNOLOGY (PART 1: UNDERSTANDING TECHNOLOGY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define technology in simple words (using tools to make work easier).
2. Mention three technological items found in the home.
3. State two ways technology helps us.
4. Identify simple tools used for communication.

CONTENT



1. What is Technology?

- **Technology means using tools, machines, and clever ideas to do things more easily, quickly, and better.**
- It is about solving problems and making life more comfortable.
- **Simple Example:** Using a **spoon** to eat soup is easier and neater than using your fingers. The spoon is a simple technology.

2. Technology in Our Homes

- We are surrounded by technology at home. These are machines or tools that help us.
 - **Television (TV):** We use it to watch cartoons, news, and learn new things.
 - **Electric Iron:** We use it to smooth our crumpled clothes and make them neat.
 - **Refrigerator (Fridge):** We use it to keep our water cold and our food fresh for many days so it doesn't spoil.
 - **Fan/Ceiling Fan:** It blows air to keep us cool when the weather is hot.
 - **Radio:** We listen to music, stories, and news on it.
 - **Blender/Mixer:** It helps Mummy to grind pepper and tomatoes for soup quickly.

3. How Technology Helps Us

- **Makes Work Easier and Faster:** Washing clothes with a washing machine is much easier and faster than washing by hand.
- **Saves Time and Energy:** A car or bus takes us to school faster and with less tiredness than walking a long distance.
- **Improves Communication:** We can talk to family far away instantly with a phone.
- **Makes Life Comfortable and Fun:** Fans keep us cool, TVs entertain us, and fridges keep our drinks cold.

4. Technology for Communication

- **Communication** means sharing information, talking, or sending messages to others.
- **Simple Communication Tools:**

- **Telephone/Mobile Phone:** We use this to call or send text messages to Daddy, Mummy, or friends who are not with us.
- **Radio and TV:** They bring news, music, and educational programs from far places to our home.
- **Letter (Post):** Writing on paper and sending it through the post office is an older technology.

EVALUATION

1. What does technology do to our work? (It makes it easier).
2. Mention two machines you have in your house.
3. What do we use to keep food fresh and cold?
4. Which technology helps us talk to people far away?
5. Is a spoon a type of simple technology?

ASSIGNMENT

1. Draw a picture of a Television.
2. Ask your parents to show you an Electric Iron and tell you what it does.
3. List three things in your living room that use electricity.

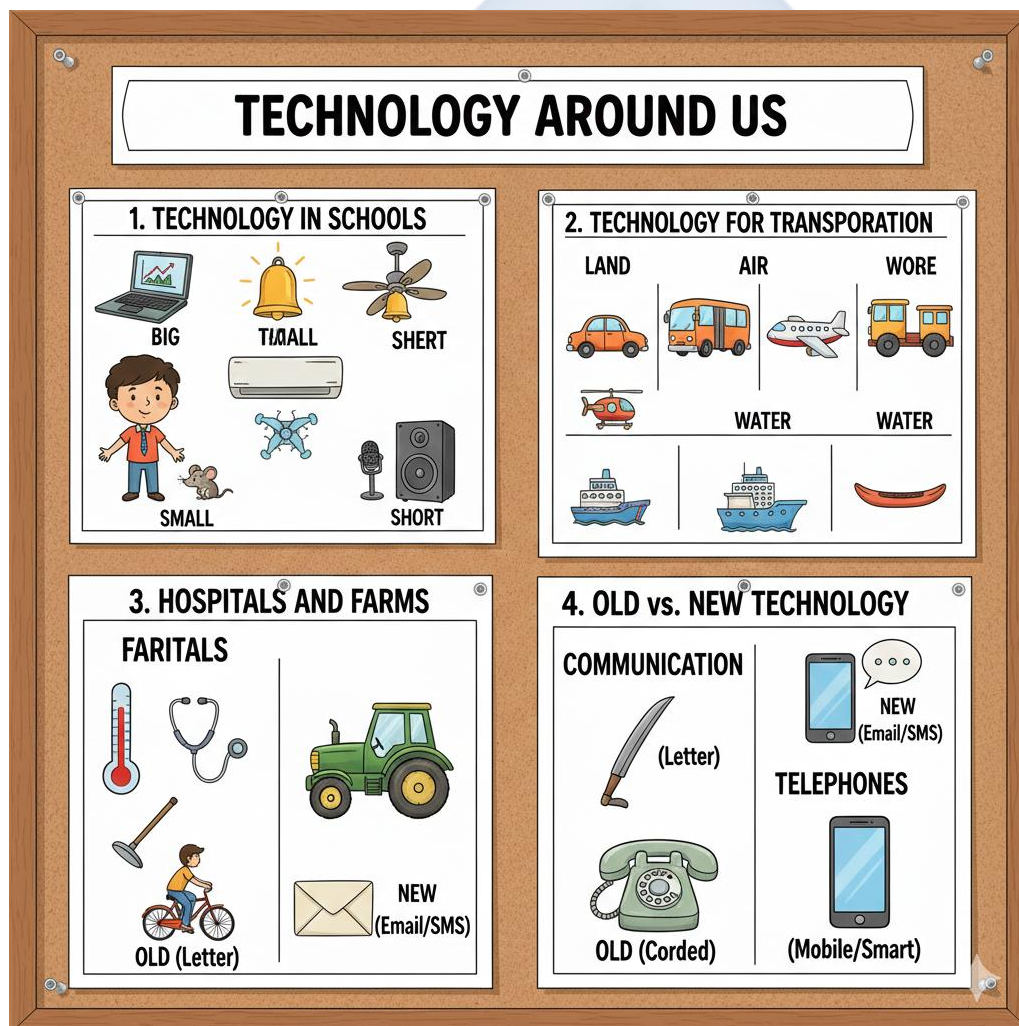
WEEK 2: TECHNOLOGY (PART 2: TECHNOLOGY AROUND US)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify three technological items found in the school.
2. List three modes of transportation (technology for moving).
3. Mention two tools used by farmers.
4. Compare an old technology with a new one (e.g., letter vs. email).

CONTENT



1. Technology in Schools

- Schools use different technologies to help teachers teach and pupils learn.
 - **Computer/Laptop:** Used for typing, drawing, playing educational games, and finding information.
 - **Electric Bell:** Rings loudly to tell us when it is time for break, assembly, or closing time.
 - **Light Bulbs/Tube Lights:** Give us bright light in the classroom so we can see the board and our books clearly, even when it's dark or cloudy.
 - **Fans/Air Conditioners:** Keep the classroom cool for better learning.
 - **Public Address System (Microphone & Speaker):** Helps the headteacher or teacher speak to many people during assembly.

2. Technology for Transportation

- **Transportation** means moving people and goods from one place to another.
- Technology has given us different **vehicles** for land, air, and water.
 - **Land Transport:** Cars, Buses, Motorcycles (Okada), Bicycles, Trains, Lorries/Trucks.
 - **Air Transport:** Aeroplanes, Helicopters, Rockets (for space).
 - **Water Transport:** Boats, Ships, Canoes, Ferries.

3. Technology in Hospitals and Farms

- **In Hospitals (Medical Technology):** Helps doctors and nurses take care of sick people.
 - **Thermometer:** A small tool used to check if your body is hot (has fever).
 - **Stethoscope:** Used by doctors to listen to your heartbeat and breathing.
 - **X-ray Machine:** Takes a picture of the inside of your body (like bones) to see if anything is broken.
- **On Farms (Agricultural Technology):** Helps farmers grow food.
 - **Hoe:** Used for digging and turning the soil.
 - **Cutlass:** Used for cutting grass and clearing bushes.
 - **Tractor:** A big machine that helps plough large farms quickly.
 - **Watering Can/Sprinkler:** Used to water plants.

4. Old vs. New Technology

- Technology keeps changing and improving over time. Newer technology is often faster, smaller, or smarter.
- **Communication:**
 - **Old:** People wrote **letters** on paper. Someone would carry the letter by foot, bicycle, or car. It took **days or weeks** to arrive.
 - **New:** We send **emails** or **text messages (SMS)** on mobile phones and smartphones. They arrive in **seconds**, no matter how far!
- **Telephones:**
 - **Old Telephones:** Big, had a wire (cord), and could only stay in one place in the house (landline).
 - **New Telephones (Mobile Phones/Smartphones):** Small, have no wires, and can go anywhere with us. We can even watch videos on them.

EVALUATION

1. Mention one machine found in the school.
2. What do we use to travel in the air?
3. What tool does a farmer use to dig?
4. What does a doctor use to check your temperature?
5. Which is faster: Writing a letter or sending a text message?

ASSIGNMENT

1. Draw a Car or a Bus.
2. Name two things a doctor uses.
3. Ask your grandparents or parents what kind of phone they used when they were young.

WEEK 3: MEASURING THINGS (PART 1: LENGTH AND SIZE)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define measurement as finding out the size or amount of something.
2. Differentiate between long and short objects.
3. Measure items using non-standard units (hand span, foot).
4. Compare the length of two objects.

CONTENT

1. Understanding Measurement

- **Measurement means finding out exactly how big, long, heavy, full, or hot something is.**
- We measure things so we can know, describe, and compare them. For example, we measure cloth to know if it's enough for a dress.

2. Measuring Length: Long and Short

- **Length** tells us **how long** an object is from one end to the other.
- **Long Objects:** A snake, a tall tree, a long ruler (30cm), a skipping rope, a stretch of road.
- **Short Objects:** A worm, a short pencil, a crayon, a nail, a paperclip.
- **Comparing Lengths:** We use words like **longer, shorter, longest, shortest**.
 - "This ruler is **longer than** that pencil."
 - "The yellow crayon is the **shortest** of all."

3. Non-Standard Units

- Long ago, before rulers and tapes, people used parts of their own body to measure things. These are called **non-standard units** because they can differ from person to person.
 - **Hand Span:** Spread your hand wide. The distance from the tip of your thumb to the tip of your pinky finger is one hand span.

- **Foot Length:** The length of your foot from heel to toe.
- **Arm Length (Cubit):** The length from your elbow to the tip of your middle finger.
- **Stride:** The length of one big step.
- **Object Units:** We can also use a pencil, a stick, or a book to measure.

4. Measuring Activities

- **Activity 1:** Use your **hand span** to measure the width of your desk. How many hand spans is it?
- **Activity 2:** Use your **foot length** to measure the length of the classroom. Walk heel-to-toe and count your steps.
- **Discussion:** Why might your friend get a different number when measuring the same table? (Because their hand span might be bigger or smaller than yours).

EVALUATION

1. What do we use to find out how long something is? (Measurement).
2. Which is longer: A new pencil or a used sharpened pencil?
3. Use your hand span to measure your desk. What is the number?
4. Who has a longer foot: You or your teacher?
5. If object A is big and object B is small, which is shorter?

ASSIGNMENT

1. Measure the length of your bed using your hand span. Write down the number.
2. Draw a long stick and a short stick.
3. Stand next to your friend. Who is taller?

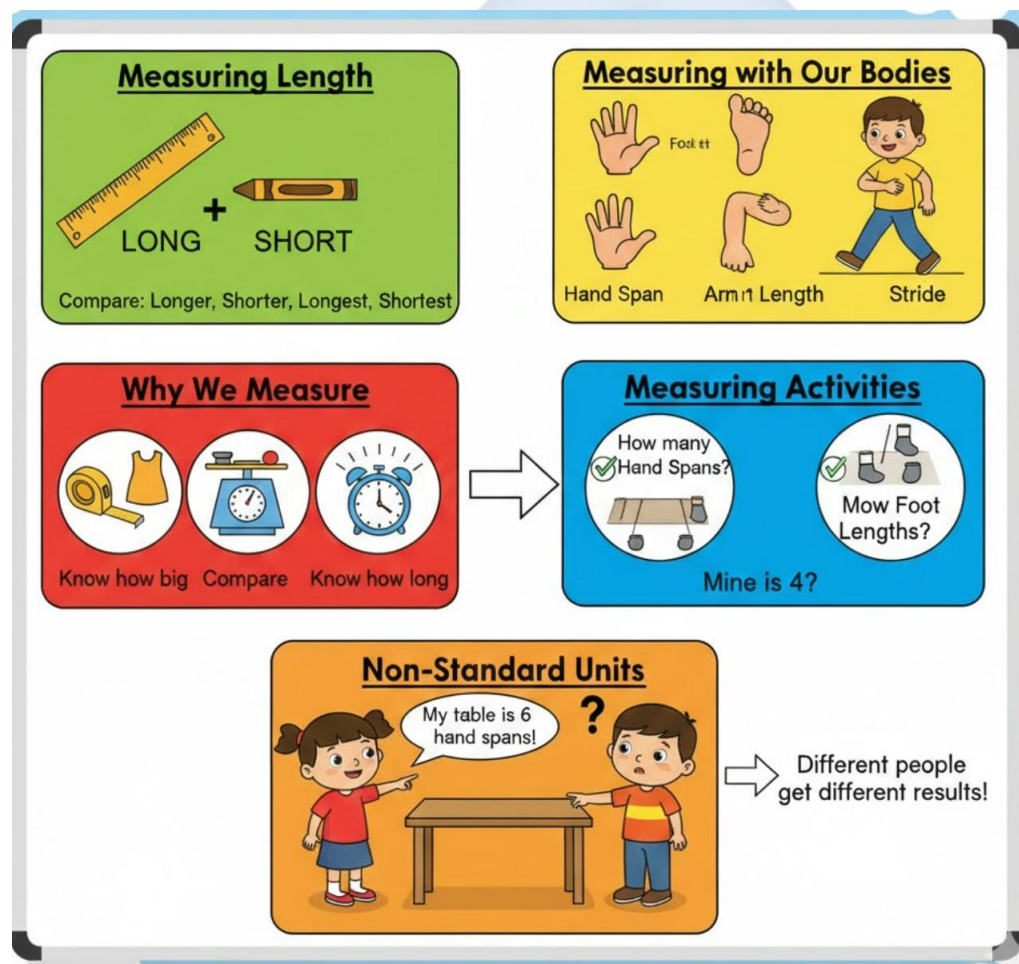
WEEK 4: MEASURING THINGS (PART 2: WEIGHT AND CAPACITY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Differentiate between heavy and light objects.
2. Identify which object is heavier using a simple balance scale.
3. Define capacity using the terms "full" and "empty."
4. Compare containers to see which holds more.

CONTENT



1. Measuring Weight: Heavy and Light

- **Weight** tells us **how heavy** something is. It is the pull of gravity on an object. We feel weight when we try to lift something.
- **Heavy Objects:** Hard to lift or carry. Examples: A big stone, a bag of rice, a car, a large water pot, a refrigerator.
- **Light Objects:** Easy to lift or carry. Examples: A feather, a piece of paper, a balloon, a cotton ball, a leaf.

2. Comparing Weights

- We can **feel and compare** weight by holding an object in each hand.
 - Example: Hold your book in one hand and your pencil in the other. The book feels **heavier**.
- **Using a Simple Balance Scale:**

[Object A] --- Beam --- [Object B]

| |
 Pan Pan

- A balance scale has two pans hanging from a beam.
- We put one object in each pan.
- The **heavier** object will make its side go **DOWN**.
- The **lighter** object will make its side go **UP**.
- If both sides are level, the objects have the **same weight**.

3. Measuring Capacity

- **Capacity** tells us **how much** a container can hold inside it (like water, sand, or rice).
- We describe capacity using these words:
 - **Full:** The container cannot hold any more. The water is at the very top.
 - **Half-full:** The container is filled halfway up.
 - **Empty:** There is nothing inside the container.
 - **Nearly Full / Nearly Empty:** Almost full or almost empty.

4. Comparing Capacity

- We can compare which container holds **more** or **less**.
 - A **bucket** holds **more** water than a **cup**.
 - A **spoon** holds **less** garri than a **bowl**.
 - A **water tank** holds the **most** water.
- We can find out by pouring water from one container to another to see which fills up first.

EVALUATION

1. Which is heavier: A stone or a leaf?
2. If a cup has no water in it, is it full or empty?
3. On a balance scale, does the heavy object go up or down?
4. Which holds more water: A bucket or a spoon?
5. Name one heavy thing in the classroom.

ASSIGNMENT

1. Lift your school bag and your water bottle. Write down which one is heavier.
2. Draw a cup that is Full and a cup that is Empty.
3. Find a bottle and a cup at home. Pour water to see which one holds more.

WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions based on the topics covered from Week 1 to Week 4.

CONTENT

Assessment of Weeks 1–4

- Understanding Technology.
- Technology around us.
- Measuring Length.
- Measuring Weight and Capacity.

EVALUATION

- Administer the Mid-Term Test.

ASSIGNMENT

- Nil.

WEEK 6: MID-TERM BREAK

OBJECTIVES

By the end of the week, pupils should be able to:

1. Rest and refresh for the second half of the term.

CONTENT

Holiday/Rest

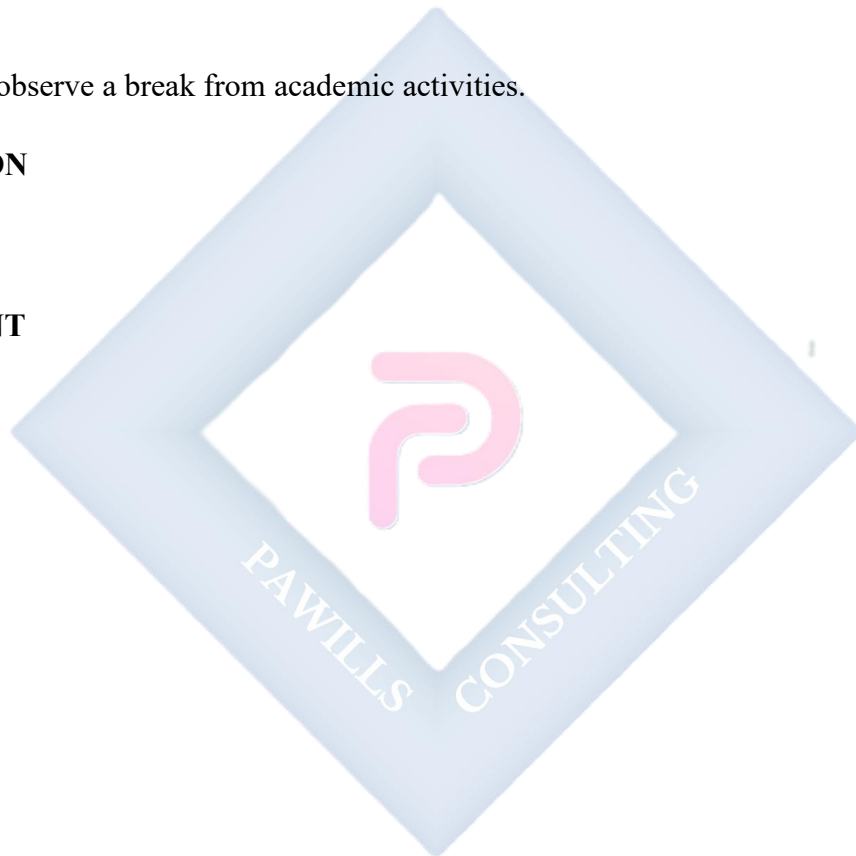
- Pupils observe a break from academic activities.

EVALUATION

- Nil.

ASSIGNMENT

- Nil.



WEEK 7: MEASURING THINGS (PART 3: TIME AND TEMPERATURE)

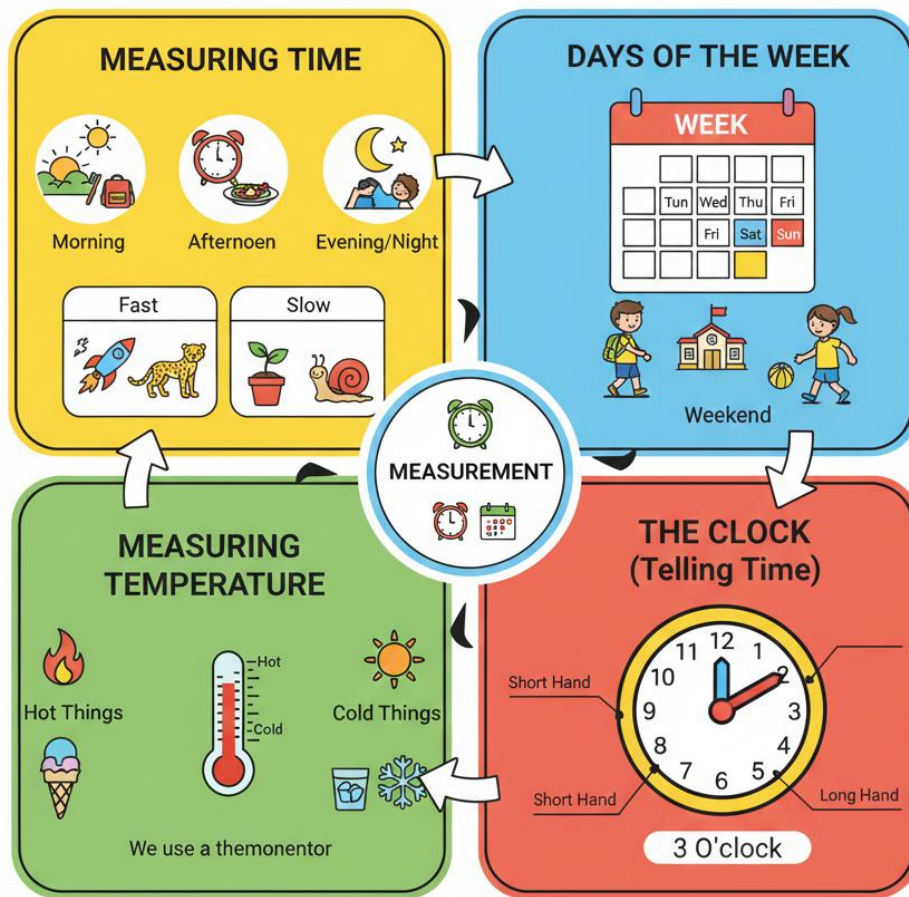
OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify parts of the day (morning, afternoon, evening).
2. Read the time as "O'clock" on a clock face.
3. Differentiate between hot and cold objects.
4. Name the days of the week.

CONTENT





1. Measuring Time

- **Time** tells us **when things happen** and **how long they take**.
- **Parts of the Day:**
 - **Morning:** From when the sun rises (around 6 am) until about 12 noon. We wake up, brush our teeth, and go to school.
 - **Afternoon:** From about 12 noon until the sun starts to go down (around 4-6 pm). We eat lunch and have afternoon lessons.
 - **Evening/Night:** From when it gets dark until we go to sleep. We eat dinner, watch TV, and sleep.

- **Fast vs. Slow:** Time can seem to pass quickly or slowly.
 - **Fast things:** An aeroplane flying, a car speeding, a cheetah running.
 - **Slow things:** A snail moving, a plant growing, the hour hand on a clock.

2. Days of the Week

- There are **7 days** in a week.
- Their order is: **Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday.**
- We go to school from **Monday to Friday**. **Saturday and Sunday** are called the weekend, often for resting and playing.

3. The Clock (Telling Time - O'Clock)

- A clock helps us know the exact time.
- A typical analog clock has:
 - **Numbers 1 to 12** in a circle.
 - A **Short Hand (Hour Hand):** It points to the hour. It moves slowly.
 - A **Long Hand (Minute Hand):** It points to the minutes. It moves faster.
- **When the Long Hand points straight up to the number 12, we say it is "O'clock".**
The time is whatever number the **Short Hand** is pointing to.
 - Example: If the short hand is on **3** and the long hand is on **12**, the time is **3 O'clock**.

4. Measuring Temperature

- **Temperature** tells us **how hot or cold** something is.
- **Hot Things:** Fire, boiling water, the sun, hot food, an electric iron when it's on.
- **Cold Things:** Ice block, ice cream, cold water from the fridge, snow, early morning air.
- **Warm Things:** Not too hot, not too cold. Just nice. Example: A cup of tea that has cooled down a bit, bath water.
- We use a **thermometer** to measure temperature exactly.

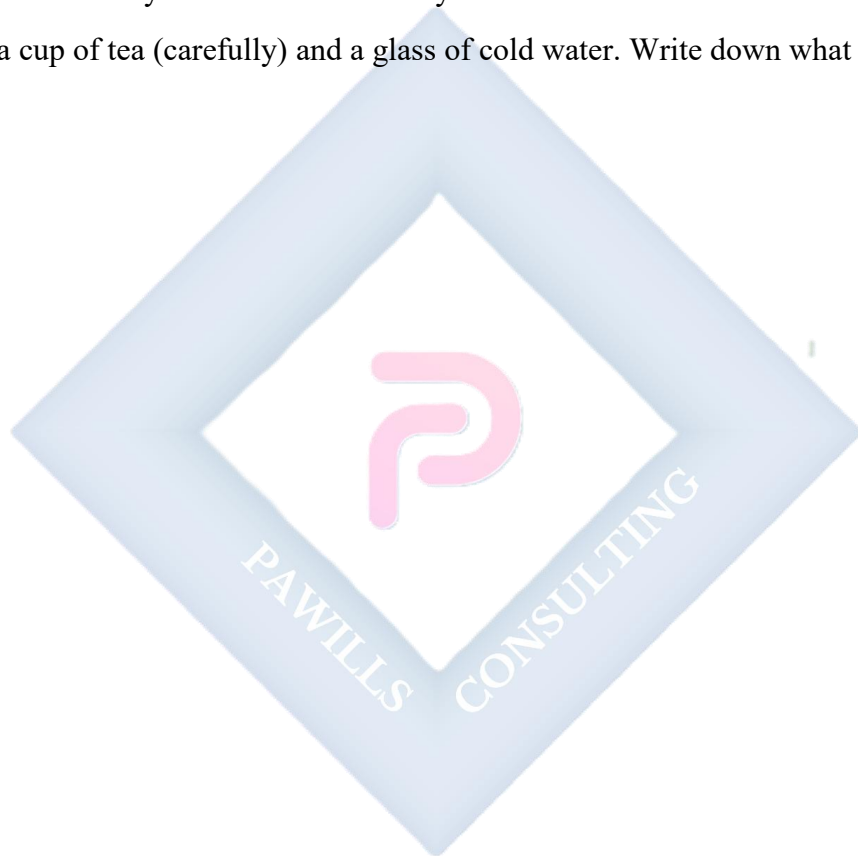
EVALUATION

1. How many days make a week?

2. What part of the day do we sleep?
3. Is fire hot or cold?
4. Is ice cream hot or cold?
5. Where is the long hand pointing when the time is "O'clock"?

ASSIGNMENT

1. Draw a clock face showing 6 O'clock.
2. Write down the days of the week correctly.
3. Touch a cup of tea (carefully) and a glass of cold water. Write down what you felt.



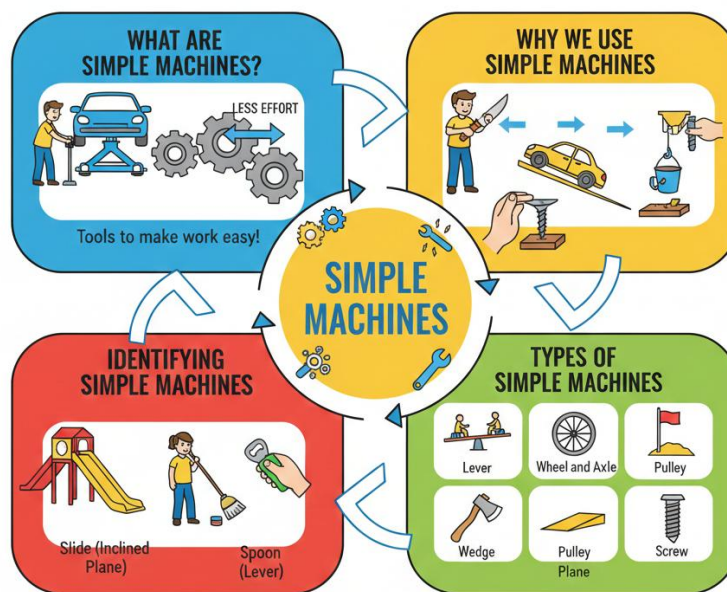
WEEK 8: SIMPLE MACHINES (PART 1: INTRODUCTION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define simple machines as tools that make work easier.
2. State two reasons why we use simple machines.
3. Identify three simple machines found in the environment.
4. List the names of common simple machines (Lever, Pulley, etc.).

CONTENT



1. What are Simple Machines?

- Simple machines are basic tools that help us do work with less effort (force). They make hard tasks feel easy.

- They have few or no moving parts.
- **Work** means moving something, lifting something, or cutting something.

2. Why We Use Simple Machines

- To **lift** heavy things (e.g., using a car jack to lift a car).
- To **cut** or split things (e.g., using scissors to cut paper).
- To **hold** things together tightly (e.g., using a screw).
- To **move** things from one place to another (e.g., using a wheelbarrow).
- To **change the direction** of a force (e.g., using a pulley to pull something up).

3. Identifying Simple Machines Simple machines are all around us! You use them every day.

- A **spoon** is a simple machine (a kind of lever) for eating.
- A **broom** is a simple machine (a kind of lever) for sweeping.
- A **slide** at the playground is a simple machine (an inclined plane).
- A **bottle opener** is a simple machine (a lever).
- A **staircase** is a simple machine (an inclined plane).

4. Types of Simple Machines There are **six main types** of simple machines:

1. **Lever** (e.g., seesaw, scissors, broom)
2. **Wheel and Axle** (e.g., bicycle, doorknob, car wheel)
3. **Inclined Plane** (e.g., ramp, slide, staircase)
4. **Pulley** (e.g., flagpole, well with a bucket, curtain rod)
5. **Wedge** (e.g., knife, axe, nail)
6. **Screw** (e.g., bottle cap, light bulb, a real screw)

EVALUATION

1. What is a simple machine?
2. Why do we use machines?
3. Is a broom a simple machine?
4. Mention two simple machines you can see in the class.

5. Does a machine make work harder or easier?

ASSIGNMENT

1. Look around your kitchen. List three tools Mummy uses to cook or clean.
2. Draw a spoon.
3. Complete the sentence: Simple machines make work _____.



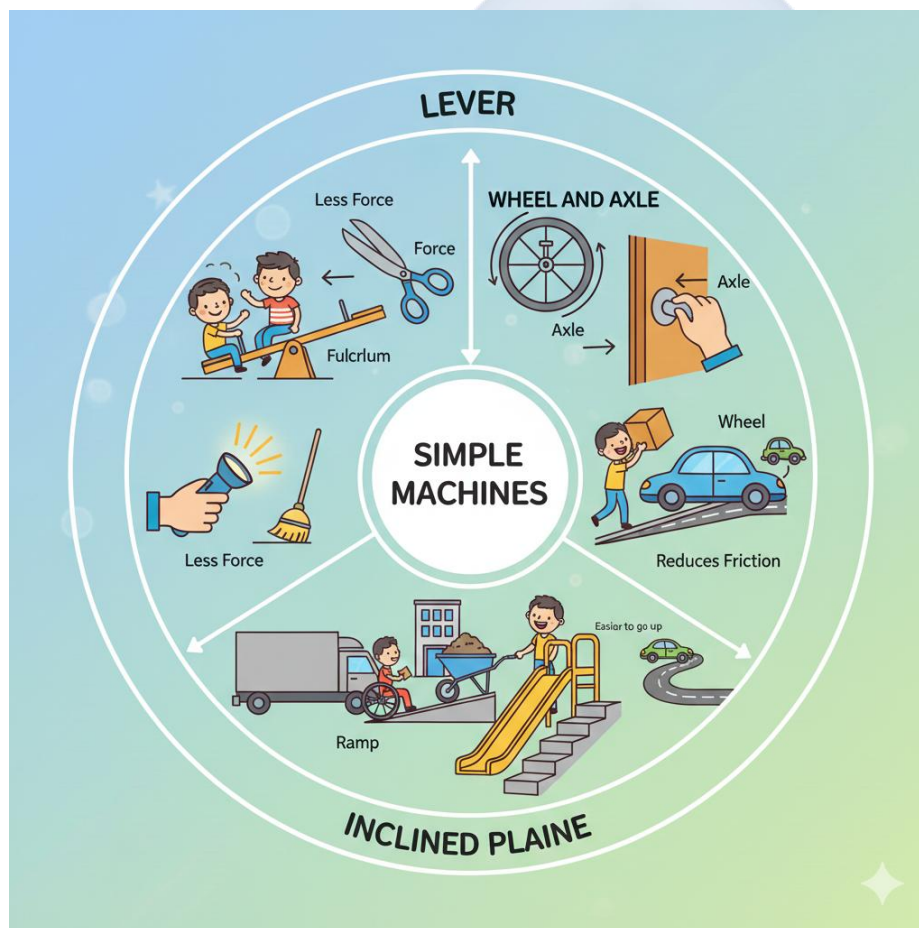
WEEK 9: SIMPLE MACHINES (PART 2: TYPES)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify examples of levers (seesaw, scissors).
2. Identify examples of the wheel and axle (bicycle, car).
3. Identify examples of an inclined plane (ramp, slide).
4. Explain how these machines help us.

CONTENT



1. Lever

- A **lever** is a stiff bar (like a plank or stick) that rests on a fixed point called a **fulcrum**. It helps to lift or move loads.

- **How it helps:** It allows you to use less force to lift something heavy.
- **Examples:**
 - **Seesaw:** The middle support is the fulcrum. Two children sit on the ends.
 - **Scissors:** The point where the two blades join is the fulcrum. You apply force at the handles to cut at the blades.
 - **Broom:** Your top hand acts as the fulcrum, your bottom hand applies force to sweep.
 - **Bottle Opener:** The tip that hooks under the cap is the fulcrum.

2. Wheel and Axle

- This is made of a **wheel** (a round object) attached to a smaller rod called an **axle** (which goes through the center of the wheel). When the wheel turns, the axle turns too.
- **How it helps:** It makes moving things very easy by reducing friction (rubbing). It is much easier to roll something than to drag it.
- **Examples:**
 - **Bicycle/Car Wheels:** The wheel turns around the axle to move the vehicle.
 - **Doorknob:** The knob (wheel) is attached to an axle that turns the latch to open the door.
 - **Wheelbarrow:** Has one wheel at the front to make it easy to push heavy loads.
 - **Toy Car:** The axles connect the wheels.

3. Inclined Plane

- An **inclined plane** is a flat surface that is slanted or sloped (not flat on the ground). It is also called a **ramp**.
- **How it helps:** It allows us to move a heavy object **up or down** more easily than lifting it straight up. You use less force, but you have to push/pull it over a longer distance.
- **Examples:**
 - **Slide:** You slide down the sloped surface for fun.
 - **Ramp:** Used for wheelchairs to enter buildings, or to push a heavy box into a truck.
 - **Staircase:** A series of small inclined planes that make it easier to walk up to a higher place.

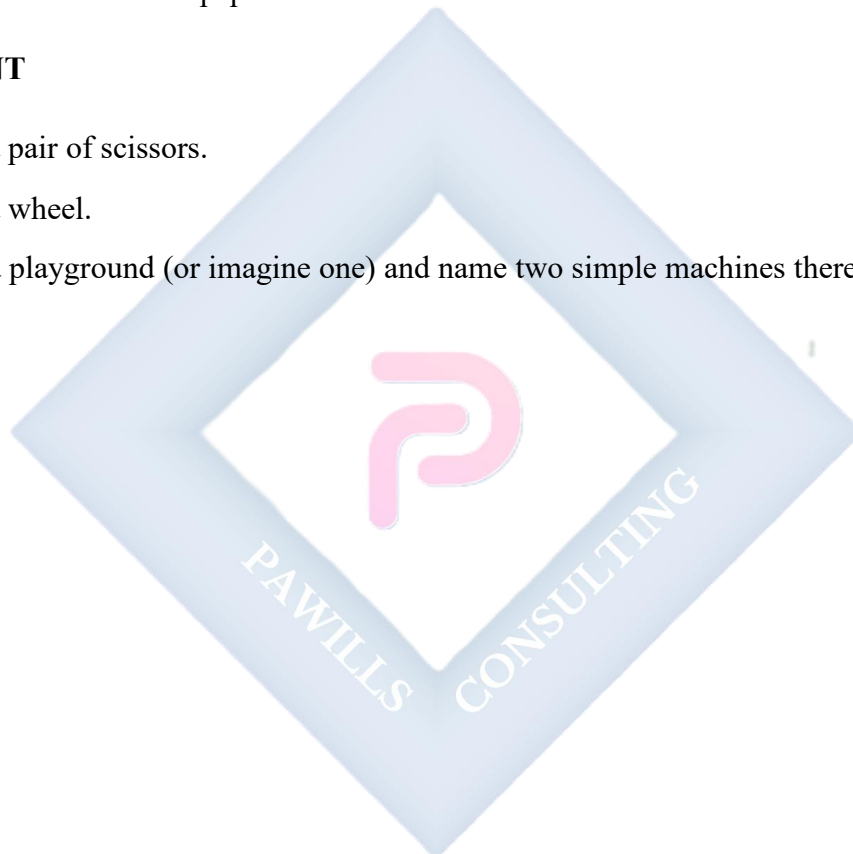
- **Road going up a hill:** A very long inclined plane for cars.

EVALUATION

1. Name one example of a lever.
2. What simple machine is on a bicycle?
3. What simple machine helps us slide down at the playground?
4. Is a doorknob a wheel and axle?
5. What do we use to cut paper?

ASSIGNMENT

1. Draw a pair of scissors.
2. Draw a wheel.
3. Go to a playground (or imagine one) and name two simple machines there.



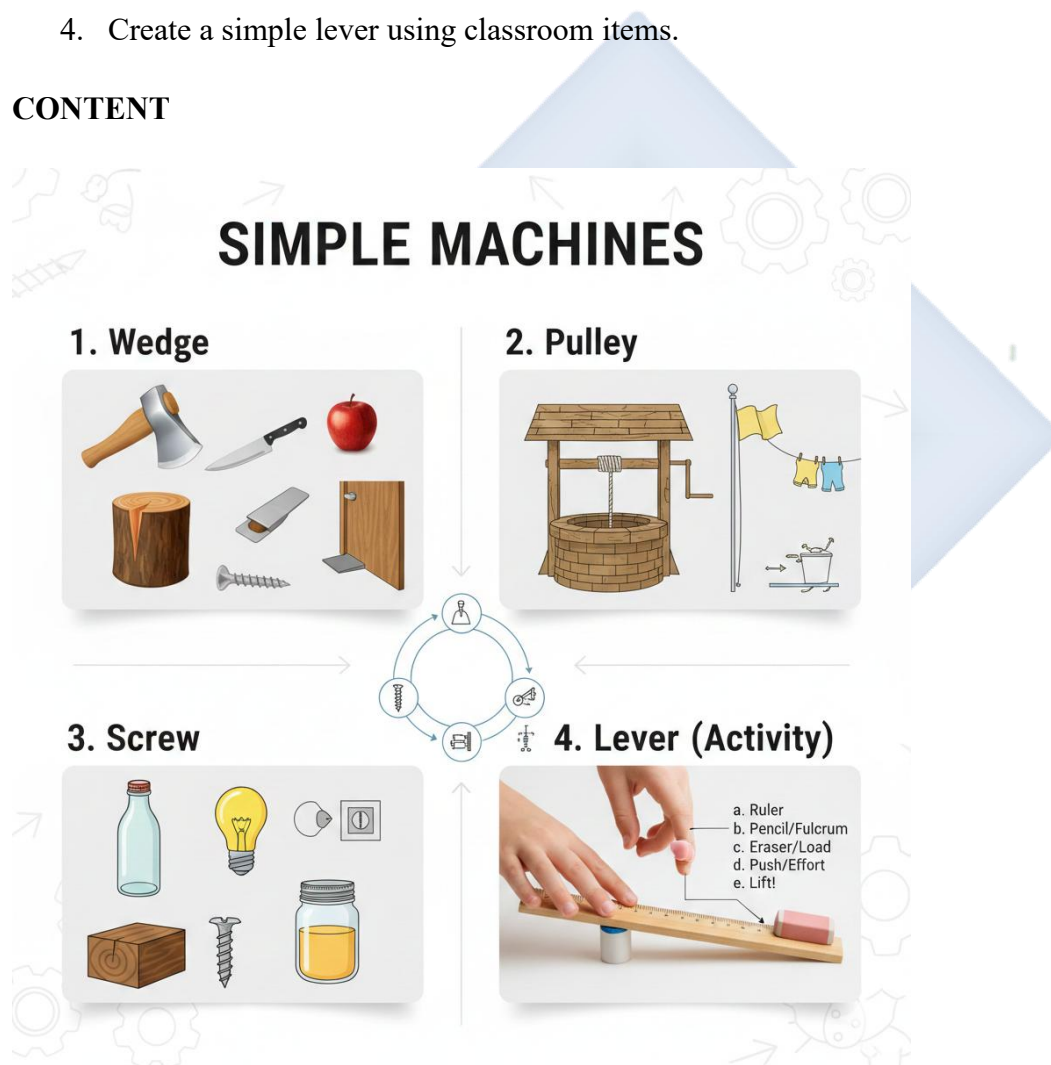
WEEK 10: SIMPLE MACHINES (PART 3: MORE TYPES AND USES)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify examples of a wedge (knife, axe).
2. Identify examples of a pulley (well bucket, flag pole).
3. Identify examples of a screw (bottle cap).
4. Create a simple lever using classroom items.

CONTENT



1. Wedge

- A **wedge** is a simple machine that is **thick at one end and thin and sharp at the other**. It is used to push things apart, cut, or hold things in place.

- **How it helps:** When you push the thick end, the thin, sharp edge applies great force to cut or split something.
- **Examples:**
 - **Knife:** The sharp blade is a wedge that cuts through food.
 - **Axe:** Used to split wood. The sharp wedge-shaped head drives into the wood.
 - **Nail:** The pointed tip is a wedge that helps it go into wood.
 - **Door Stopper:** A wedge pushed under a door to hold it open.
 - **Scissors/Blades:** The cutting edges are wedges.

2. Pulley

- A **pulley** is a **wheel with a groove (a cut) around it**, with a rope or chain running through the groove. It changes the direction of the force needed to lift something.
- **How it helps:** It makes lifting heavy loads upwards easier. You pull **down** on the rope to make the load go **up**.
- **Examples:**
 - **Well with a Bucket:** The rope goes **over** a pulley wheel. You pull down on one end of the rope, and the bucket of water comes up from the well.
 - **Flag Pole:** The rope goes through a pulley at the top. You pull down on one end to raise the flag up.
 - **Clothesline (with pulley):** Some lines use pulleys to move the clothes along.
 - **Window Blinds/Curtain Rods:** Often use a pulley system to open and close them.

3. Screw

- A **screw** is an **inclined plane wrapped around a cylinder (a pole)**. It has ridges called **threads** that go around in a spiral.
- **How it helps:** It holds things together very tightly. When you turn (rotate) a screw, the threads pull it into the material, creating a strong grip.
- **Examples:**
 - **Bottle Cap/Screw Cap:** The inside of the cap has threads that match the threads on the bottle neck. Turning it screws it on or off.

- **Light Bulb:** The metal base has threads that screw into the socket.
- **Wood/Metal Screws:** Used to join pieces of wood or metal together. They have a pointed tip and threads.
- **Jar Lid:** Works like a bottle cap.

4. Making a Simple Machine (Lever)

- **Activity: Make a Mini Seesaw**
 - a. Find a **ruler** (the lever/bar).
 - b. Place a **pencil or small cylindrical eraser** under the ruler's middle point (this is the **fulcrum**).
 - c. Place a **small eraser** on one end of the ruler (this is the **load**).
 - d. Gently push down on the other end of the ruler (this is your **effort**).
 - e. Watch the eraser (load) lift up! You have made a **lever**.

EVALUATION

1. What simple machine is a knife? (Wedge).
2. What helps us pull water from a well? (Pulley).
3. What holds a bottle cap tight? (Screw).
4. Mention one object that is a wedge.
5. What happens when you use a pulley to pull a flag? (It goes up).

ASSIGNMENT

1. Draw a Knife (Wedge).
2. Draw a Bottle Cap (Screw).
3. Find a screw in your house (look at a toy or a chair) and tell your parents you found a simple machine.

WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from Technology, Measurement, and Simple Machines.
2. Demonstrate readiness for the end-of-term examination.

CONTENT

Review of All Topics

- **Technology:** Meaning, examples at home (TV, fridge, iron) and school (computer, bell), transportation, communication (old vs new).
- **Measurement:**
 - **Length:** Long vs short, measuring with hand span/foot.
 - **Weight:** Heavy vs light, using a balance scale.
 - **Capacity:** Full, half-full, empty. Which holds more?
 - **Time:** Parts of day, days of week, reading "O'clock" on a clock.
 - **Temperature:** Hot, cold, warm.
- **Simple Machines:** Six types: Lever, Wheel & Axle, Inclined Plane, Pulley, Wedge, Screw. Examples of each and how they help.

EVALUATION

- Oral quiz identifying simple machines in pictures (e.g., show a picture of a seesaw, knife, ramp).
- Practical measurement tasks (e.g., "Who is taller?" "Is this bag heavier than this book?").
- Naming parts of a computer or home appliances.

ASSIGNMENT

- Study all notes from Week 1 to Week 10 in preparation for the examination.